NM-W4-Bodorin

Numerical solution of algebraic and transcendental equations

The bisection method · The secant method

Numerical Methods · Course 4 · theory step by step and 10 fully worked examples

Roadmap of the presentation

0 Setting the stage

- The problem f(x) = 0: separate, then refine
- Course example: separating the root of $x^3 - x^2 - x - 1$ with Rolle's sequence

I The bisection method

- One step: three cases
- Nested intervals and their properties
- Theorem, error bound, number of steps
- 5 worked examples

II The secant method

- Hypotheses and the 4 shape cases
- Fixed endpoint rule, formula derivation
- Theorem, error estimate, stopping
- 5 worked examples (same equations)

Comparison at the end: the same five equations solved by both methods, iteration counts side by side.

The problem: approximate a real root ξ of f(x) = 0

1 Equation

f(x) = 0, f algebraic (polynomial) or transcendental (e^x, ln x, sin x, …)

2 Root

ξ with f(ξ) = 0; usually it cannot be written in closed form

3 Approximation

a number ξ′ with $|\xi' - \xi| \le \varepsilon$ for a prescribed tolerance ε

Two phases

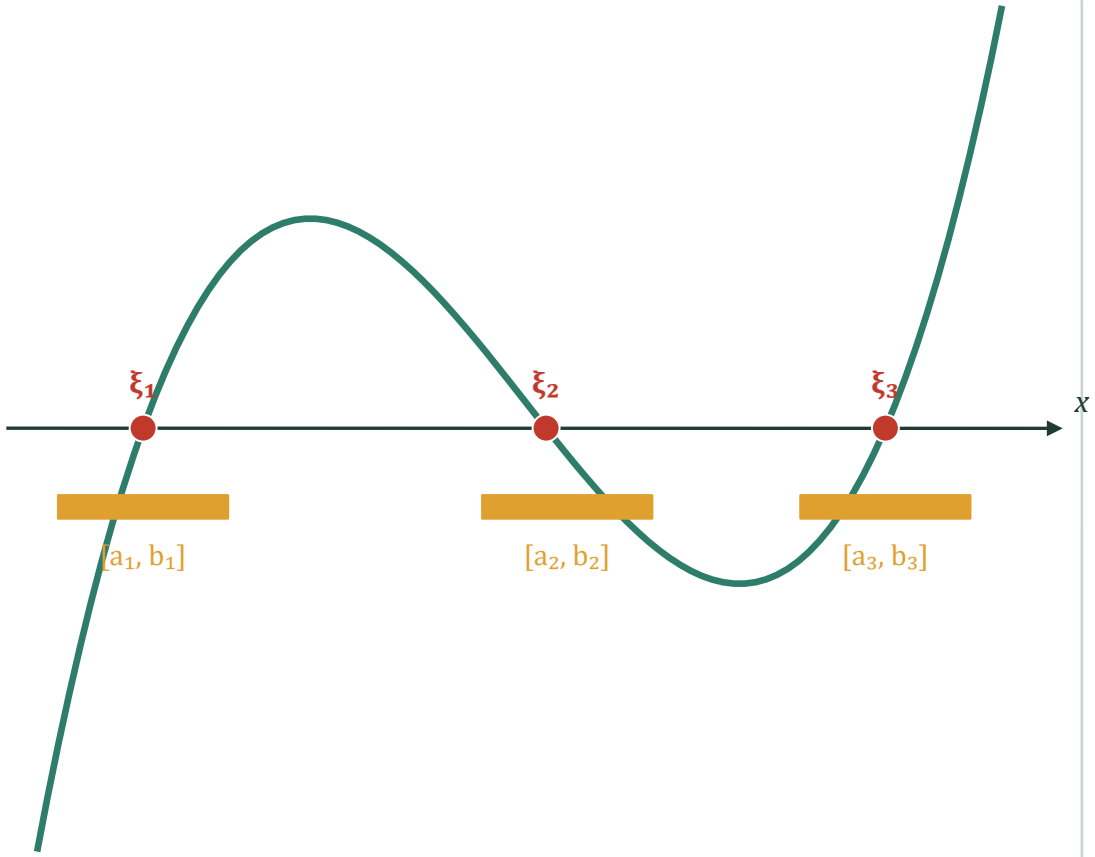
Phase 1 · Separation

find intervals [a, b] containing exactly one root
(Rolle's sequence, sign table, graph)

→

Phase 2 · Refinement

shrink the interval / iterate towards ξ
(bisection, secant, Newton, …)



Phase 1 gives the gold intervals, one root in each; this course is about Phase 2.

Course example: separate the roots of $x^3 - x^2 - x - 1 = 0$

1 Derivative

$f'(x) = 3x^2 - 2x - 1 = 0 \Rightarrow x_1 = -1/3,\ x_2 = 1$

2 Values

$f(-\infty) = -\infty,\ f(-1/3) = -22/27,\ f(1) = -2,\ f(+\infty) = +\infty$

3 Rolle's sequence

signs $-\ -\ -\ +\ \Rightarrow$ exactly one sign change

4 Conclusion

one real root $\xi \in (1, +\infty)$

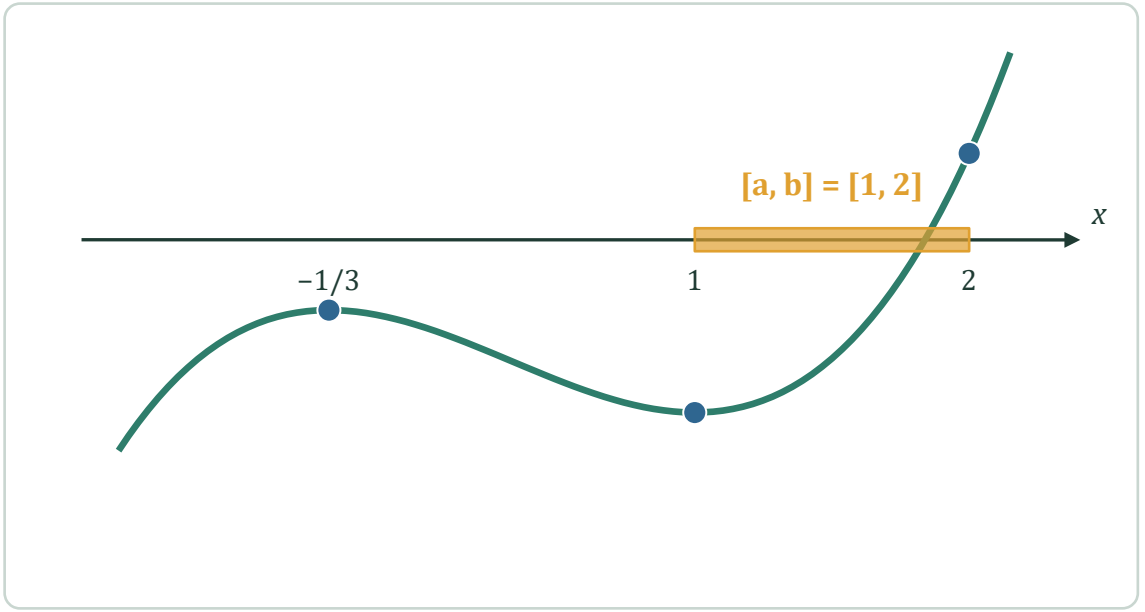
5 Narrow it

$f(1) = -2 < 0,\ f(2) = 8 - 4 - 2 - 1 = 1 > 0 \Rightarrow \xi \in (1, 2)$

This interval is the starting point for both methods (Examples B1 and S1).

x	$-\infty$	$-1/3$	1	$+\infty$
f(x)	$-\infty$	$-22/27$	-2	$+\infty$
sign	−	−	− →	+

ξ



Part I · The bisection method

Also called bipartition or interval halving: cut the interval in two, keep the half where f changes sign

Starting point and key idea of bisection

1 **f continuous on [a, b]**

no jumps, so the graph cannot skip the x-axis

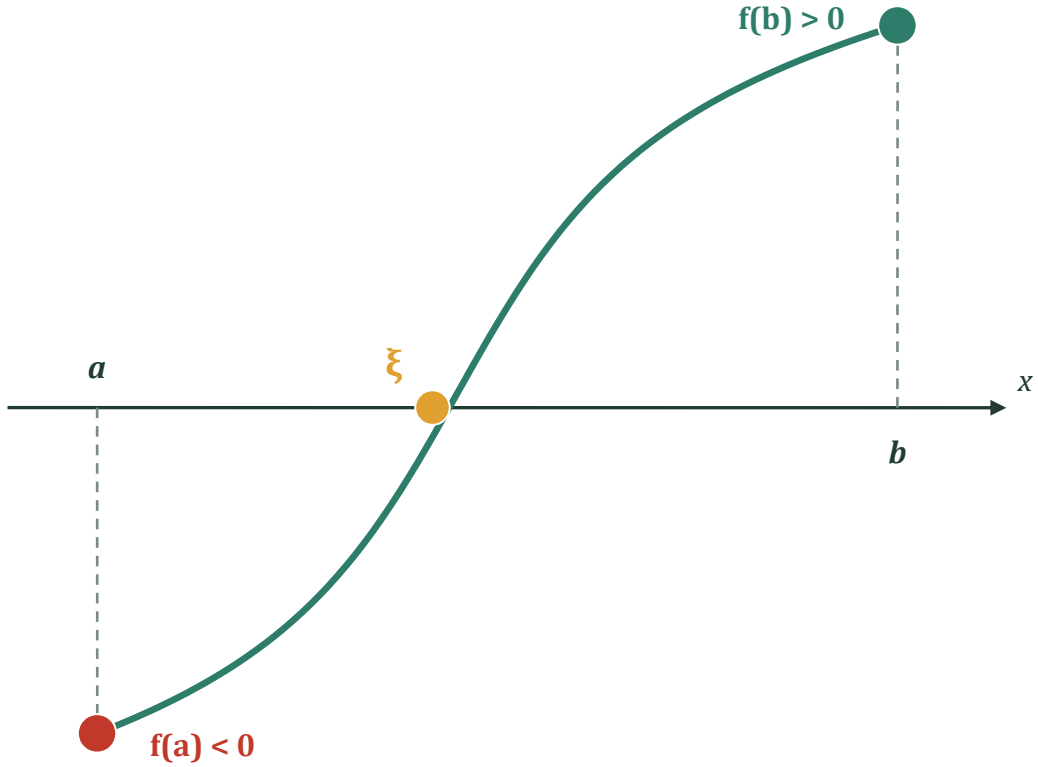
2 **f(a) · f(b) < 0**

f has opposite signs at the two ends

3 **exactly one root ξ in (a, b)**

ensured beforehand by root separation (Phase 1)

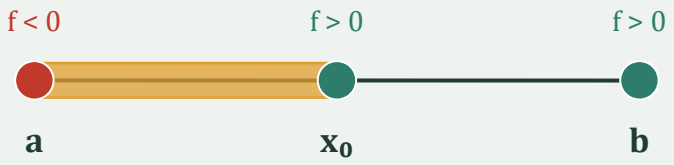
Why a root exists (Bolzano / intermediate value theorem)
A continuous function that is negative at one end and positive at the other must take the value 0 in between. Bisection uses only this fact: it needs the signs of f, never its derivatives.



One bisection step: compute the midpoint, then three cases

$I_0 = [a_0, b_0] = [a, b], \quad x_0 = (a_0 + b_0)/2 = (a + b)/2 \quad \rightarrow$ look at the sign of $f(x_0)$

a) $f(a) \cdot f(x_0) < 0$

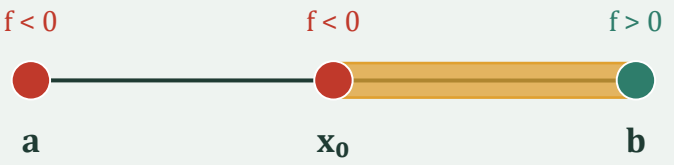


gold = kept half (sign change inside)

$a_1 = a, \; b_1 = x_0$

$I_1 = [a, x_0], \; \xi \in (a, x_0)$

b) $f(x_0) \cdot f(b) < 0$

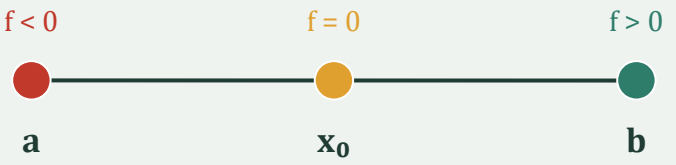


gold = kept half (sign change inside)

$a_1 = x_0, \; b_1 = b$

$I_1 = [x_0, b], \; \xi \in (x_0, b)$

c) $f(x_0) = 0$

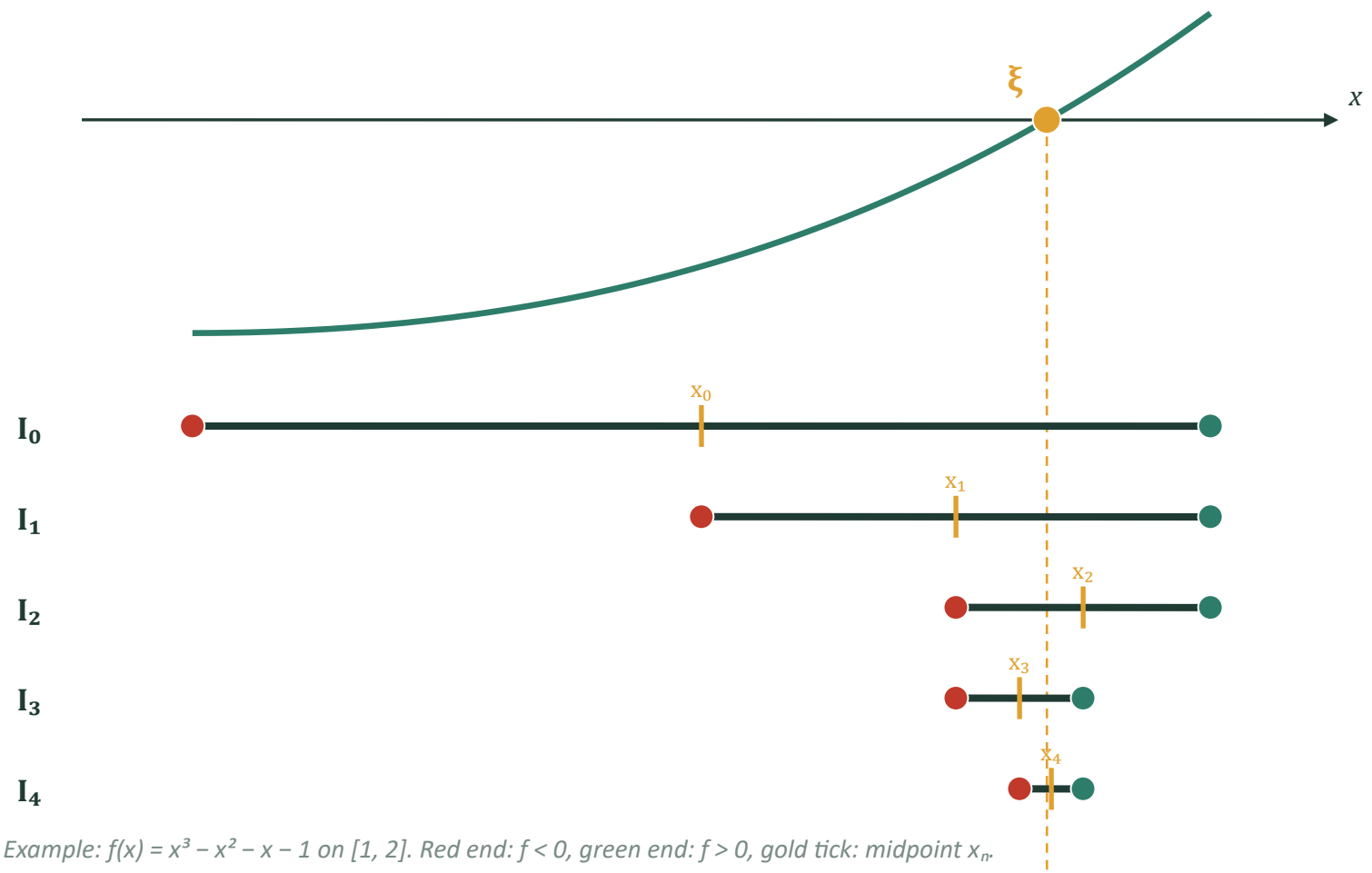


no interval is kept

$\xi = x_0 = (a + b)/2$

stop: the root is found exactly

Repeating the step: a chain of nested intervals



Example: f(x) = x³ − x² − x − 1 on [1, 2]. Red end: f < 0, green end: f > 0, gold tick: midpoint x_n.

What we obtain

$$I_0 \supset I_1 \supset I_2 \supset \ldots \supset I_n \supset \ldots$$

$I_n = [a_n, b_n]$
$x_n = (a_n + b_n)/2$

Each interval is one half of the previous one, and ξ always stays inside.

Properties of the sequences (a_n) and (b_n)

1 **Ordering** — *each step moves only one end, towards the inside*

$$a = a_0 \le a_1 \le \dots \le a_n \le \dots \le b_n \le \dots \le b_1 \le b_0 = b$$

2 **Monotone and bounded** — *⇒ both sequences converge*

(a_n) increasing, bounded above by b; (b_n) decreasing, bounded below by a

3 **Sign change kept** — *cases a) and b) are chosen exactly to keep it*

$f(a_n) \cdot f(b_n) < 0$ for every n $\Rightarrow$ $\xi \in I_n = [a_n, b_n]$

4 **Length halves** — *induction: one halving per step*

$$b_n - a_n = (b_{n-1} - a_{n-1})/2 = (b_{n-2} - a_{n-2})/2^2 = \dots = (b - a)/2^n$$

Theorem: convergence and error bound of bisection

1) $\lim a_n = \lim b_n = \xi \quad (n \to \infty)$

2) with $x_n = (a_n + b_n)/2$: $\quad |x_n - \xi| \le (b - a)/2^n \quad$ for every $n \in \mathbb{N}$

1 Limits exist

$a_n \to \alpha$ and $b_n \to \beta$ (monotone + bounded)

2 They are equal

$\beta - \alpha = \lim (b_n - a_n) = \lim (b - a)/2^n = 0 \Rightarrow \alpha = \beta$

3 The common limit is ξ

$f(a_n) \cdot f(b_n) < 0 \to$ (continuity) $f(\alpha)^2 \le 0 \Rightarrow f(\alpha) = 0$; ξ is unique $\Rightarrow \alpha = \xi$

4 Error bound

x_n and ξ both lie in $[a_n, b_n] \Rightarrow |x_n - \xi| \le b_n - a_n = (b - a)/2^n$

Sharper, for free

x_n is the midpoint of $[a_n, b_n]$, so in fact $|x_n - \xi| \le (b - a)/2^{n+1}$.

The course states the simpler bound $(b - a)/2^n$; both are guaranteed bounds, known before computing anything.

When to stop, and how many steps are needed

Stopping criteria (stop at the first n that satisfies the chosen one)

$\lvert f(x_n)\rvert < \varepsilon$	the residual is small
$b_n - a_n < \varepsilon$	the bracket is short (absolute error)
$(b_n - a_n)/\lvert x_n\rvert \le \varepsilon,\ x_n \neq 0$	relative error, useful when ξ is large or tiny

Number of steps, known in advance

want $(b - a)/2^n \le \varepsilon$

$\Leftrightarrow\ 2^n \ge (b - a)/\varepsilon$

$\Leftrightarrow\ \mathbf{n \ge \log_2((b - a)/\varepsilon)}$

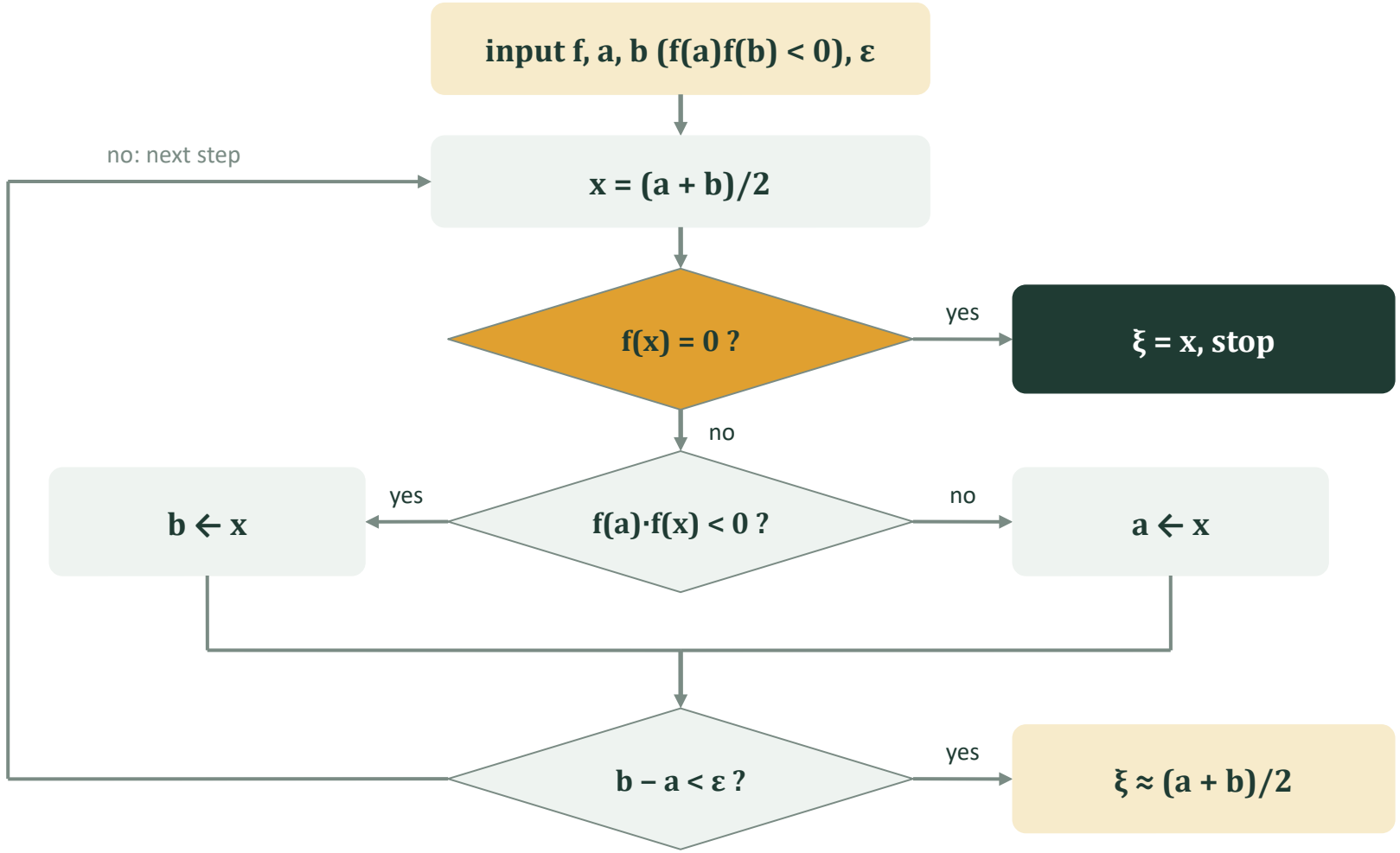
each extra correct decimal costs $\log_2 10 \approx 3.3$ more steps

b − a = 1

ε	$\log_2(1/\varepsilon)$	steps n
10^{-1}	3.32	4
10^{-2}	6.64	7
10^{-3}	9.97	10
10^{-4}	13.29	14
10^{-6}	19.93	20
10^{-8}	26.58	27

Fixed, predictable cost: the count depends only on b − a and ε, not on f.

The bisection algorithm as a flowchart



Pseudocode

```
repeat
  x ← (a + b)/2
  if f(x) = 0:
    return x
  if f(a)·f(x) < 0:
    b ← x
  else:
    a ← x
until b − a < ε
return (a + b)/2
```

Bisection in practice: strengths and limits

+ Strengths

- Always converges once $f(a)f(b) < 0$ and f is continuous
- Needs only signs of f: no derivatives
- Error bound known before starting: $(b - a)/2^n$
- Ideal for a rough first approximation of a real root

– Limits

- Slow: the error only halves per step (linear convergence, ratio 1/2)
- High precision ⇒ many steps ⇒ long computing time
- Ignores the values of f: a tiny $|f(a)|$ is not exploited
- Needs a sign change: double roots (f touches the axis) are missed

Course remark: bisection is advantageous for a coarse approximation; when very high precision is needed, the growing number of iterations limits its use.

Example B1 · bisection for $x^3 - x^2 - x - 1 = 0$: check and plan

1 **Continuity**

$f(x) = x^3 - x^2 - x - 1$ is continuous on $[1, 2]$

2 **Sign change**

$f(1) = -2.0000,\ f(2) = 1.0000 \Rightarrow f(a)\cdot f(b) < 0$ ✓

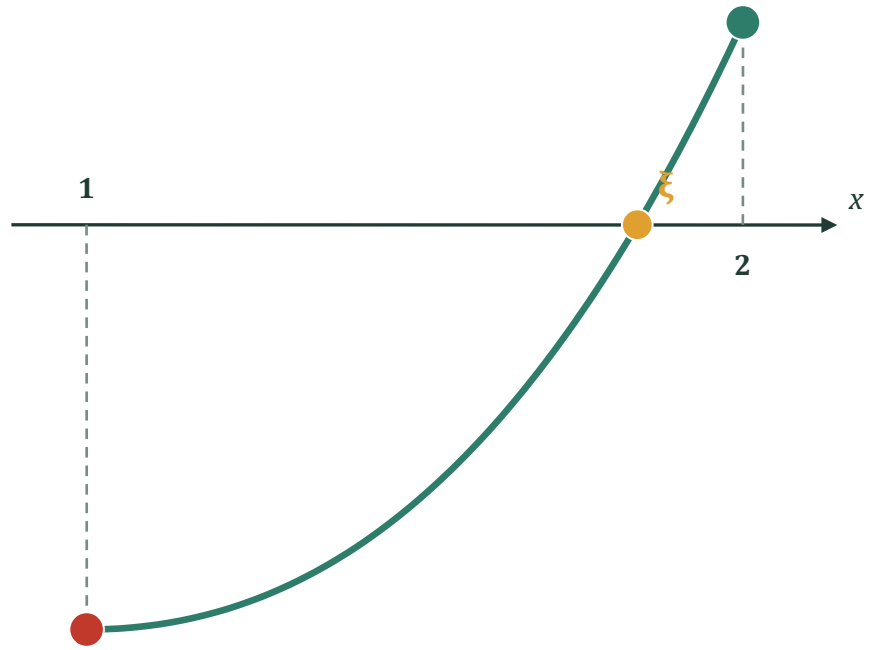
3 **One root only**

$f'(x) = 3x^2 - 2x - 1 = (3x + 1)(x - 1)$; $f' > 0$ on $(1, 2]$ ($f'(1) = 0$ only at the end) $\Rightarrow$ f strictly monotone

4 **Steps needed ($\varepsilon = 0.01$)**

$n \geq \log_2((b - a)/\varepsilon) = \log_2(100) = 6.64 \Rightarrow n = 7$

Plan: compute the midpoints $x_0, x_1, \ldots, x_7$; then $|x_7 - \xi| \leq (b - a)/2^7 = 0.0078 \leq 0.01$



Source: course example.

Example B1 · the 8 bisection steps

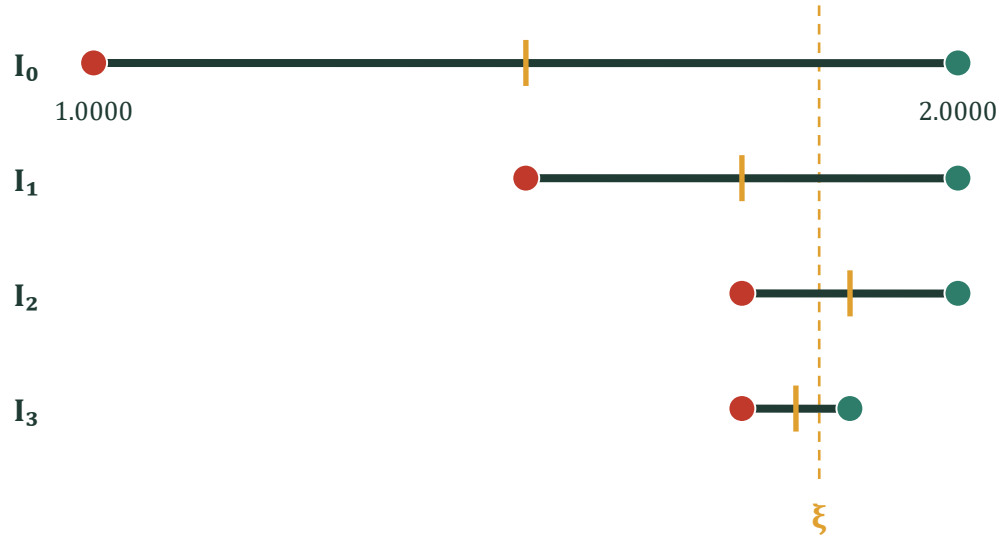
n	a_n	b_n	$x_n = (a_n + b_n)/2$	$f(a_n)$	$f(x_n)$	case and next interval	$(b - a)/2^n$
0	1.000000	2.000000	**1.500000**	–	**−1.37500**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	1.0000
1	1.500000	2.000000	**1.750000**	–	**−0.45313**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.5000
2	1.750000	2.000000	**1.875000**	–	**0.20117**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.2500
3	1.750000	1.875000	**1.812500**	–	**−0.14331**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.1250
4	1.812500	1.875000	**1.843750**	–	**0.02451**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0625
5	1.812500	1.843750	**1.828125**	–	**−0.06050**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0313
6	1.828125	1.843750	**1.835938**	–	**−0.01827**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0156
7	1.835938	1.843750	**1.839844**	–	**0.00305**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0078

How to read a row: compute x_n, evaluate $f(x_n)$, compare its sign with $f(a_n)$; the case column gives $[a_{n+1}, b_{n+1}]$ of the next row.

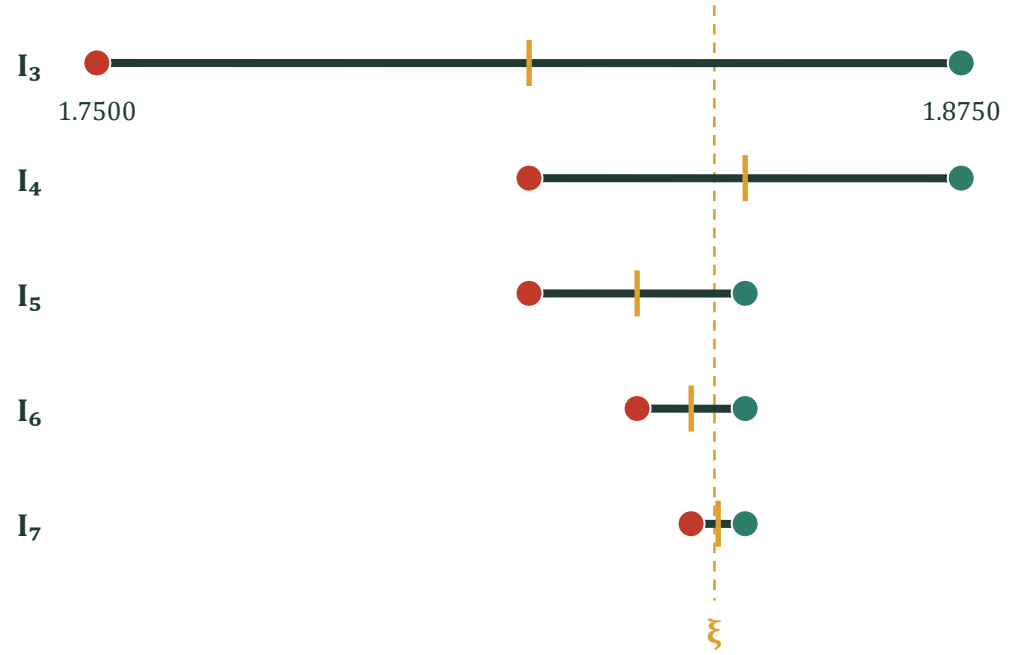
Stop at n = 7: $(b - a)/2^7 = 0.0078 \le \varepsilon = 0.01 \Rightarrow \xi \approx x_7 = 1.839844$

Example B1 · the nested intervals, drawn to scale

Full view: I_0 … I_3 on [1, 2]



Zoom ×8: I_3 … I_7 on [1.7500, 1.8750]



$\xi \approx x_7 = 1.839844$ exact $\xi = 1.83928676$

true error $|x_7 - \xi| = 5.57\times10^{-4} \le$ guaranteed bound 0.0078

Example B2 · bisection for $x^3 + x - 1 = 0$: check and plan

1 Continuity

$f(x) = x^3 + x - 1$ is continuous on $[0.5, 1]$

2 Sign change

$f(0.5) = -0.3750,\ f(1) = 1.0000 \Rightarrow f(a)\cdot f(b) < 0$ ✓

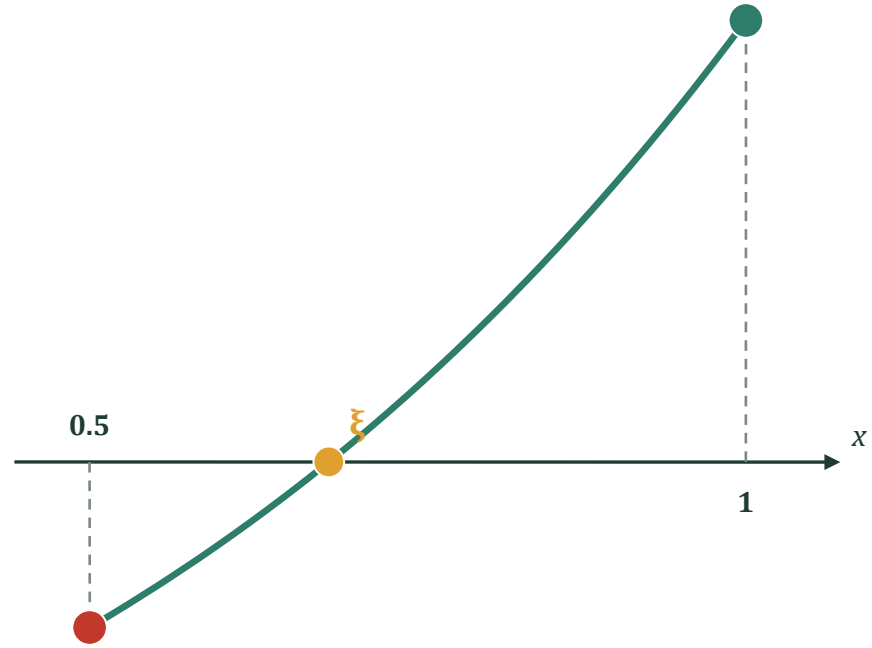
3 One root only

$f'(x) = 3x^2 + 1;\ f' \geq 1.75 > 0$ on $[0.5, 1] \Rightarrow f$ strictly monotone

4 Steps needed ($\varepsilon = 0.01$)

$n \geq \log_2((b - a)/\varepsilon) = \log_2(50) = 5.64 \Rightarrow n = 6$

Plan: compute the midpoints $x_0, x_1, \ldots, x_6$; then $|x_6 - \xi| \leq (b - a)/2^6 = 0.0078 \leq 0.01$



Source: algebraic.

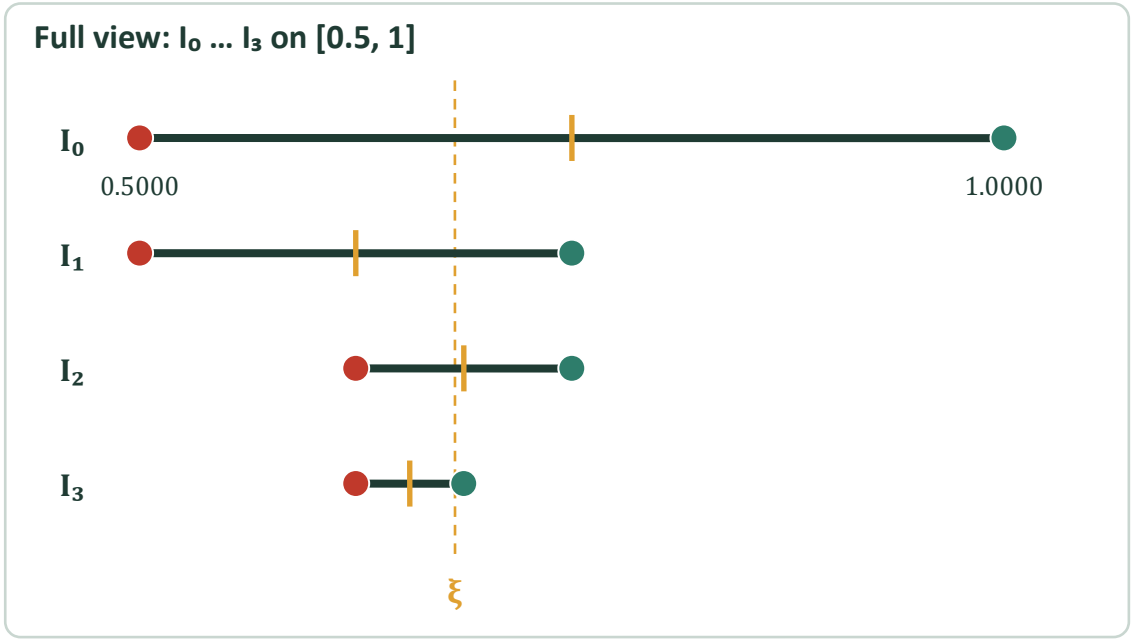
Example B2 · the 7 bisection steps

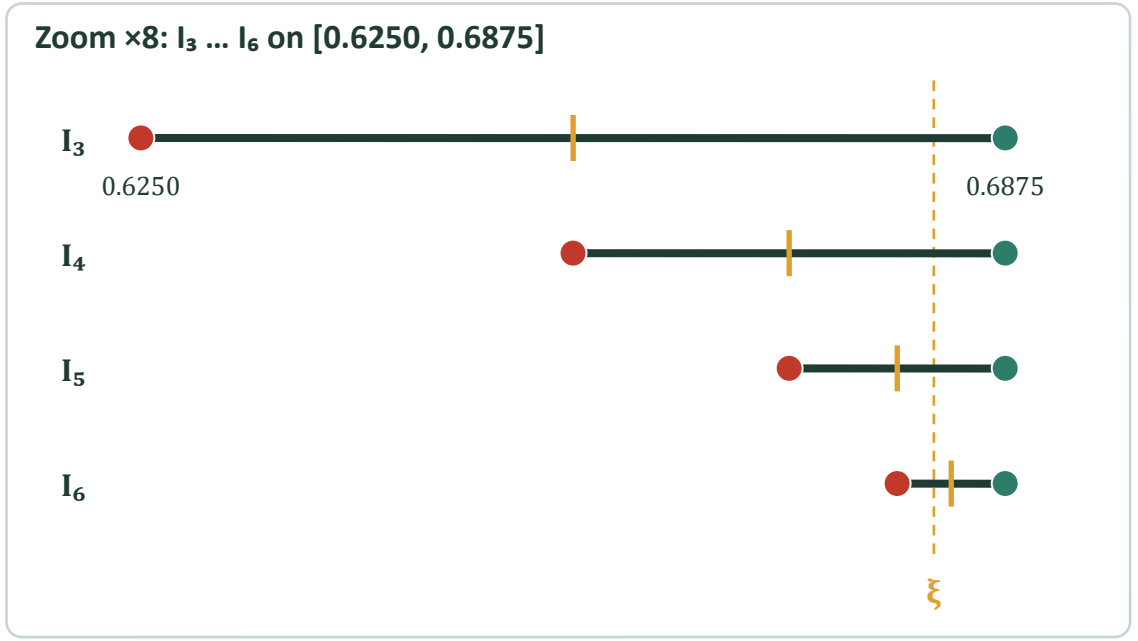
n	a_n	b_n	$x_n = (a_n + b_n)/2$	$f(a_n)$	$f(x_n)$	case and next interval	$(b - a)/2^n$
0	0.500000	1.000000	**0.750000**	–	0.17188	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.5000
1	0.500000	0.750000	**0.625000**	–	−0.13086	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.2500
2	0.625000	0.750000	**0.687500**	–	0.01245	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.1250
3	0.625000	0.687500	**0.656250**	–	−0.06113	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0625
4	0.656250	0.687500	**0.671875**	–	−0.02483	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0313
5	0.671875	0.687500	**0.679688**	–	−0.00631	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0156
6	0.679688	0.687500	**0.683594**	–	0.00304	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0078

How to read a row: compute x_n, evaluate $f(x_n)$, compare its sign with $f(a_n)$; the case column gives $[a_{n+1}, b_{n+1}]$ of the next row.

Stop at n = 6: $(b - a)/2^6 = 0.0078 \le \varepsilon = 0.01 \Rightarrow \xi \approx x_6 = 0.683594$

Example B2 · the nested intervals, drawn to scale





$\xi \approx x_6 = 0.683594$ exact $\xi = 0.68232780$

true error $|x_6 - \xi| = 1.27\times10^{-3} \le$ guaranteed bound 0.0078

Example B3 · bisection for cos x − x = 0: check and plan

1 Continuity

$f(x) = \cos x - x$ is continuous on $[0, 1]$

2 Sign change

$f(0) = 1.0000,\ f(1) = -0.4597 \Rightarrow f(a)\cdot f(b) < 0$ ✓

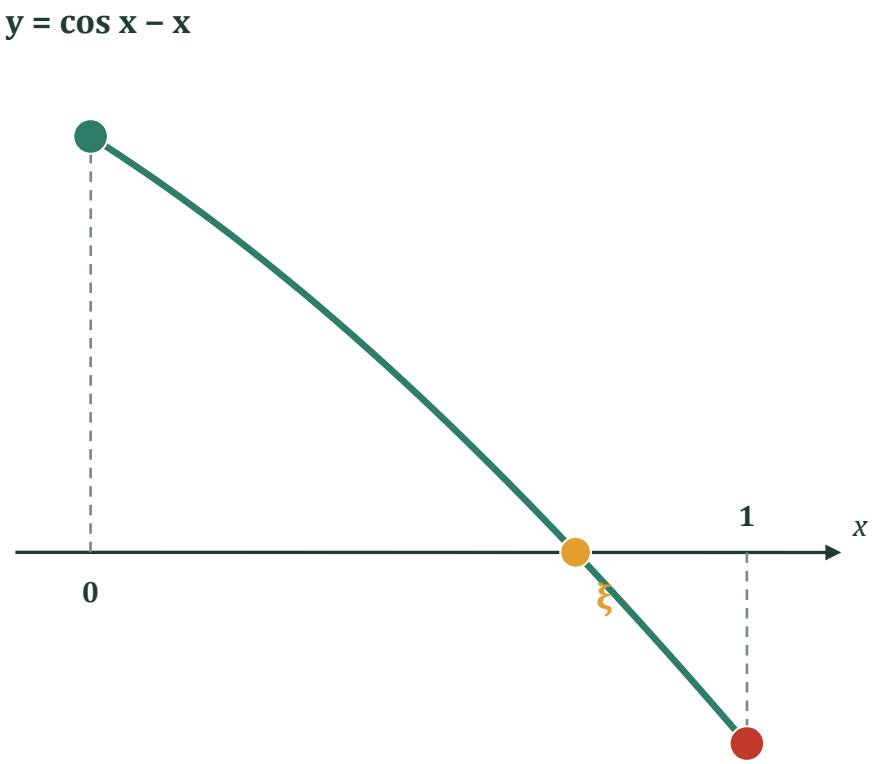
3 One root only

$f'(x) = -\sin x - 1;\ f' \le -1 < 0$ on $[0, 1] \Rightarrow$ f strictly monotone

4 Steps needed (ε = 0.01)

$n \ge \log_2((b - a)/\varepsilon) = \log_2(100) = 6.64 \Rightarrow n = 7$

Plan: compute the midpoints $x_0, x_1, \ldots, x_7$; then $|x_7 - \xi| \le (b - a)/2^7 = 0.0078 \le 0.01$



Source: transcendental.

Example B3 · the 8 bisection steps

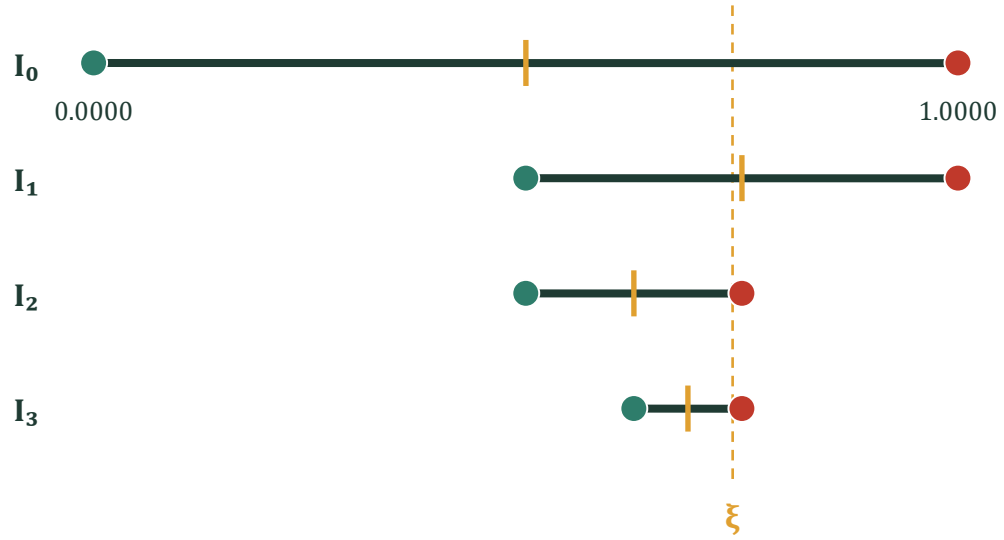
n	a_n	b_n	$x_n = (a_n + b_n)/2$	$f(a_n)$	$f(x_n)$	case and next interval	$(b - a)/2^n$
0	0.000000	1.000000	**0.500000**	+	**0.37758**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	1.0000
1	0.500000	1.000000	**0.750000**	+	**−0.01831**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.5000
2	0.500000	0.750000	**0.625000**	+	**0.18596**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.2500
3	0.625000	0.750000	**0.687500**	+	**0.08533**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.1250
4	0.687500	0.750000	**0.718750**	+	**0.03388**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0625
5	0.718750	0.750000	**0.734375**	+	**0.00787**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0313
6	0.734375	0.750000	**0.742188**	+	**−0.00520**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0156
7	0.734375	0.742188	**0.738281**	+	**0.00135**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0078

How to read a row: compute x_n, evaluate $f(x_n)$, compare its sign with $f(a_n)$; the case column gives $[a_{n+1}, b_{n+1}]$ of the next row.

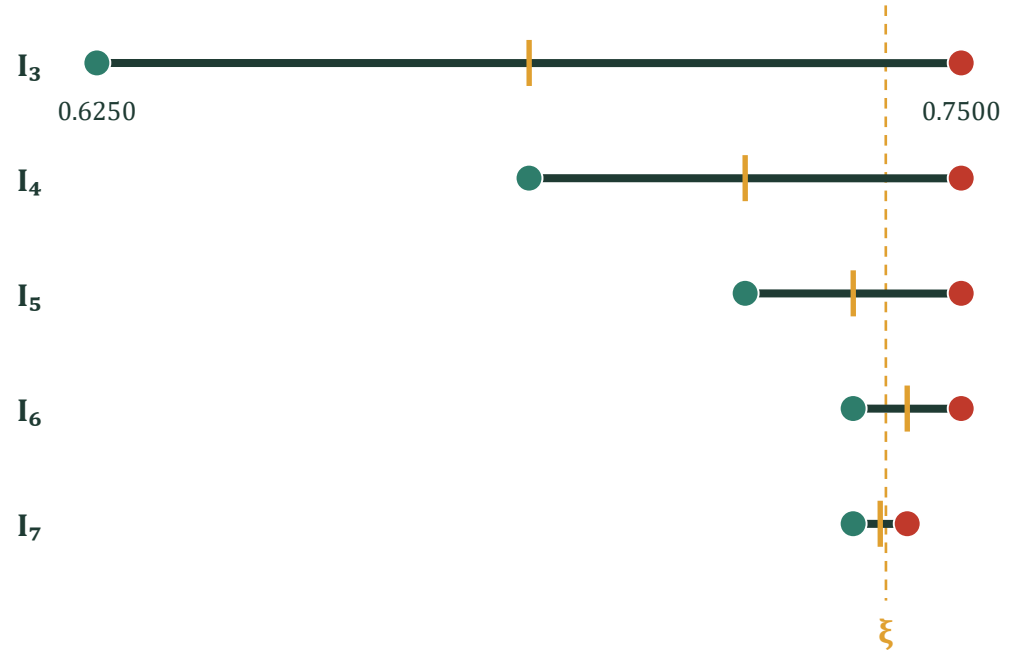
Stop at n = 7: $(b - a)/2^7 = 0.0078 \le \varepsilon = 0.01 \Rightarrow \xi \approx x_7 = 0.738281$

Example B3 · the nested intervals, drawn to scale

Full view: I_0 … I_3 on [0, 1]



Zoom ×8: I_3 … I_7 on [0.6250, 0.7500]



$\xi \approx x_7 = 0.738281$ exact $\xi = 0.73908513$

true error $|x_7 - \xi| = 8.04\times10^{-4} \le$ guaranteed bound 0.0078

Example B4 · bisection for $e^{-x} - x = 0$: check and plan

1 Continuity

$f(x) = e^{-x} - x$ is continuous on $[0, 1]$

2 Sign change

$f(0) = 1.0000,\ f(1) = -0.6321 \Rightarrow f(a)\cdot f(b) < 0$ ✓

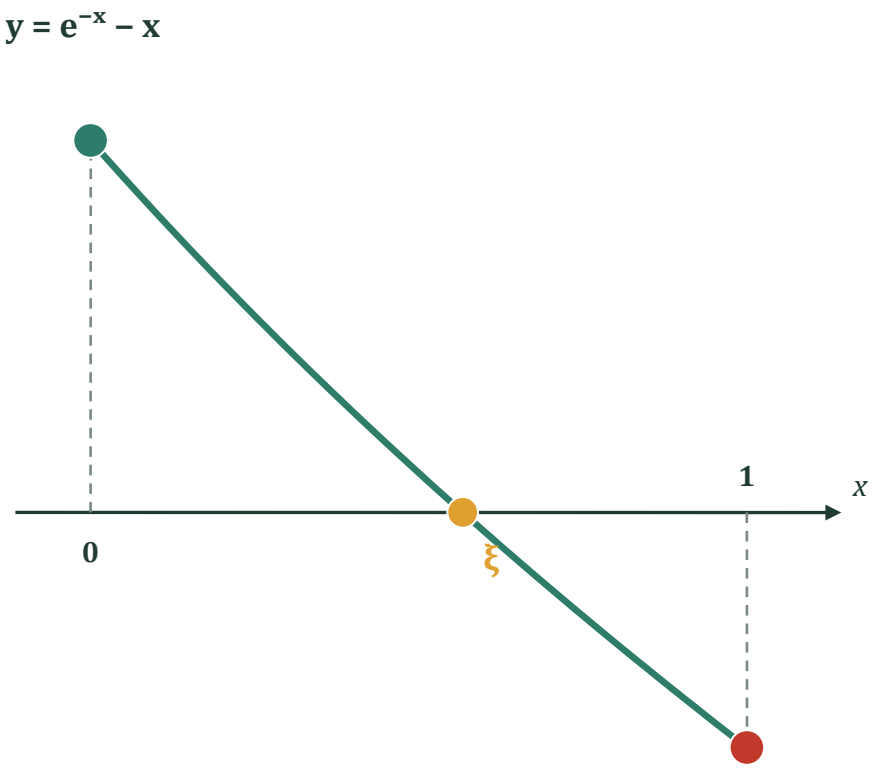
3 One root only

$f'(x) = -e^{-x} - 1;\ f' < -1 < 0$ on $[0, 1] \Rightarrow$ f strictly monotone

4 Steps needed (ε = 0.01)

$n \geq \log_2((b - a)/\varepsilon) = \log_2(100) = 6.64 \Rightarrow n = 7$

Plan: compute the midpoints $x_0, x_1, \ldots, x_7$; then $|x_7 - \xi| \leq (b - a)/2^7 = 0.0078 \leq 0.01$



Source: transcendental.

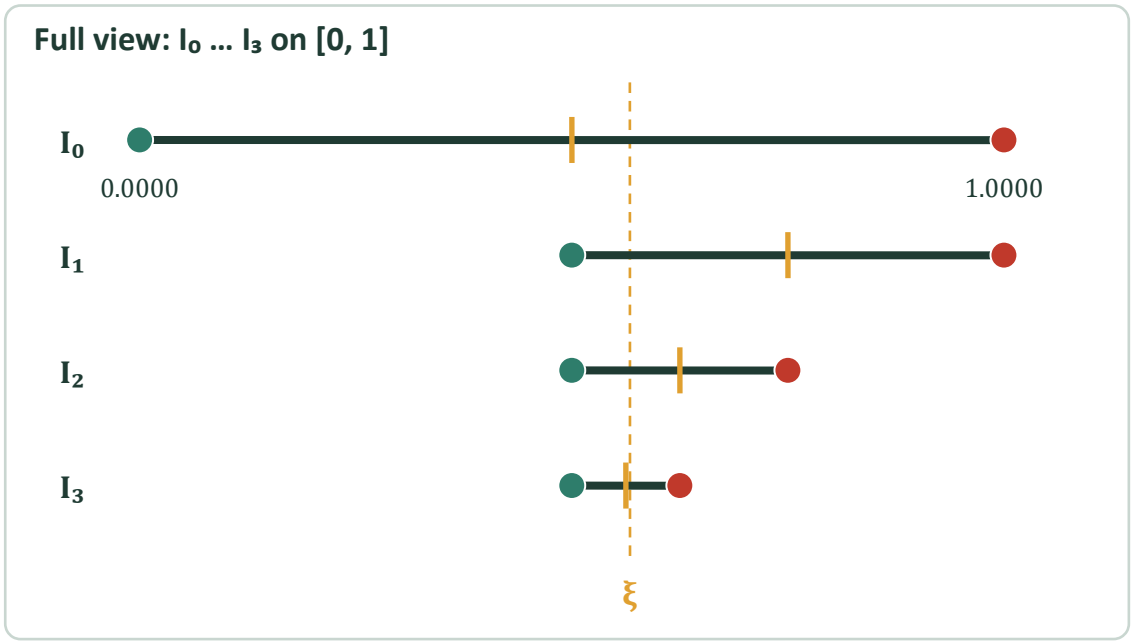
Example B4 · the 8 bisection steps

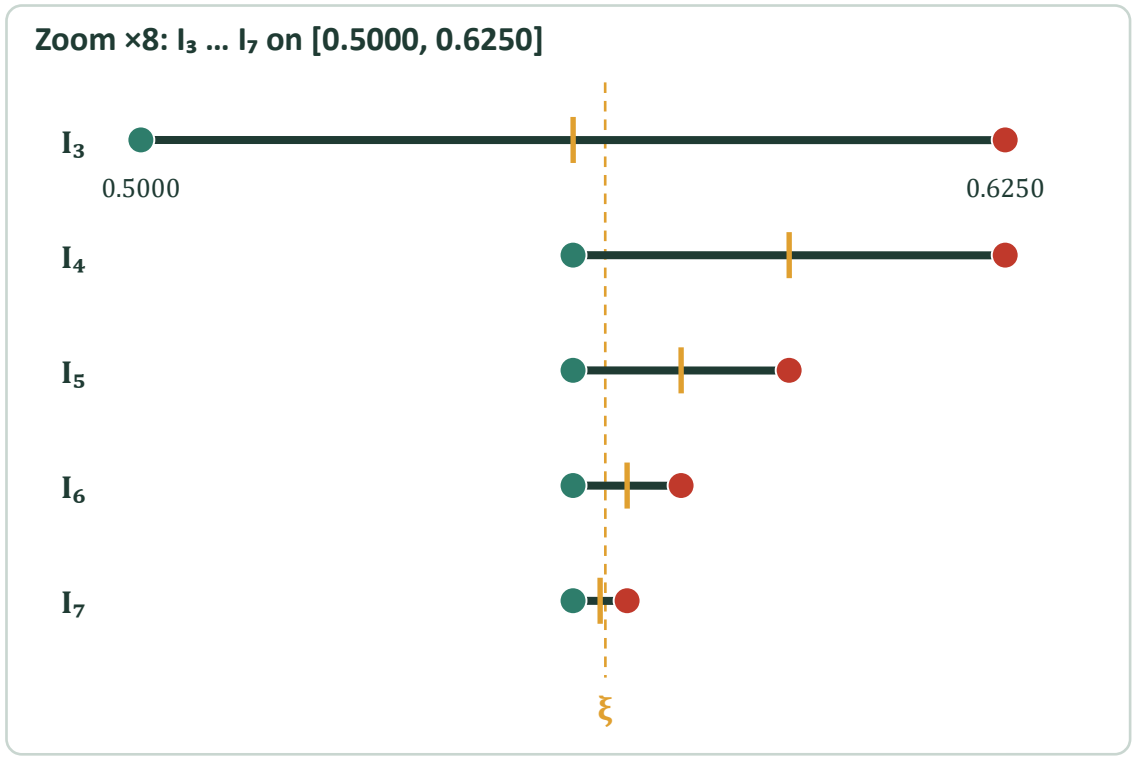
n	a_n	b_n	$x_n = (a_n + b_n)/2$	$f(a_n)$	$f(x_n)$	case and next interval	$(b - a)/2^n$
0	0.000000	1.000000	**0.500000**	+	0.10653	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	1.0000
1	0.500000	1.000000	**0.750000**	+	−0.27763	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.5000
2	0.500000	0.750000	**0.625000**	+	−0.08974	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.2500
3	0.500000	0.625000	**0.562500**	+	0.00728	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.1250
4	0.562500	0.625000	**0.593750**	+	−0.04150	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0625
5	0.562500	0.593750	**0.578125**	+	−0.01718	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0313
6	0.562500	0.578125	**0.570313**	+	−0.00496	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0156
7	0.562500	0.570313	**0.566406**	+	0.00116	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0078

How to read a row: compute x_n, evaluate $f(x_n)$, compare its sign with $f(a_n)$; the case column gives $[a_{n+1}, b_{n+1}]$ of the next row.

Stop at n = 7: $(b - a)/2^7 = 0.0078 \le \varepsilon = 0.01 \Rightarrow \xi \approx x_7 = 0.566406$

Example B4 · the nested intervals, drawn to scale





$\xi \approx x_7 = 0.566406$ exact $\xi = 0.56714329$

true error $|x_7 - \xi| = 7.37\times10^{-4} \le$ guaranteed bound 0.0078

Example B5 · bisection for ln x + x − 2 = 0: check and plan

1 **Continuity**

$f(x) = \ln x + x - 2$ is continuous on $[1, 2]$

2 **Sign change**

$f(1) = -1.0000,\ f(2) = 0.6931 \Rightarrow f(a)\cdot f(b) < 0$ ✓

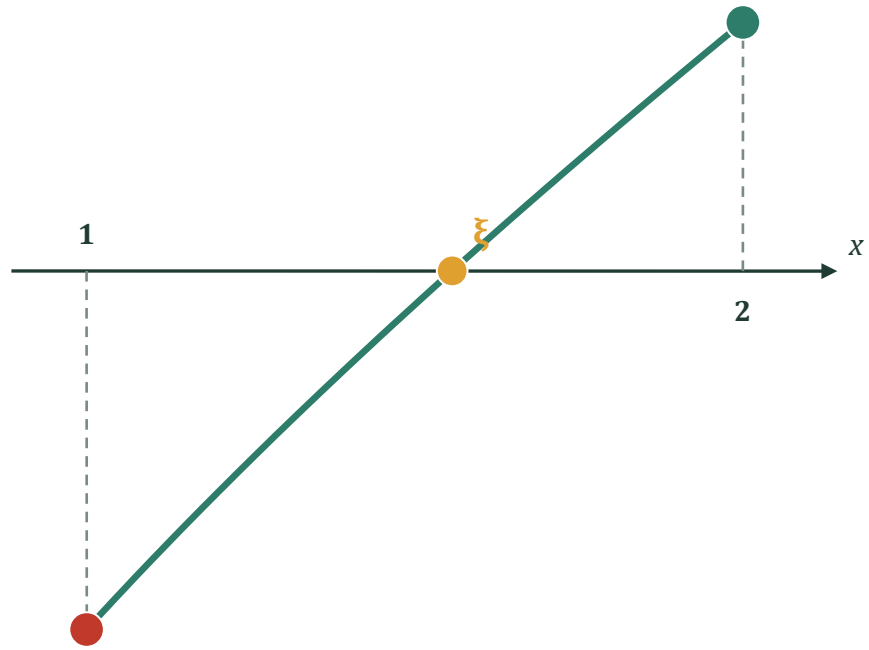
3 **One root only**

$f'(x) = 1/x + 1;\ f' > 1 > 0$ on $[1, 2] \Rightarrow$ f strictly monotone

4 **Steps needed (ε = 0.01)**

$n \geq \log_2((b - a)/\varepsilon) = \log_2(100) = 6.64 \Rightarrow n = 7$

Plan: compute the midpoints $x_0, x_1, \ldots, x_7$; then $|x_7 - \xi| \leq (b - a)/2^7 = 0.0078 \leq 0.01$



Source: transcendental.

Example B5 · the 8 bisection steps

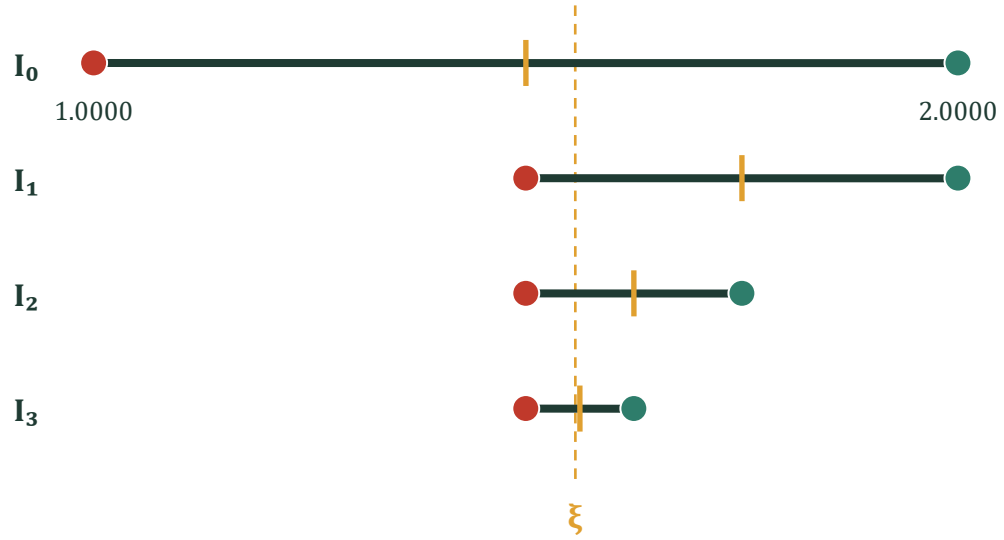
n	a_n	b_n	$x_n = (a_n + b_n)/2$	$f(a_n)$	$f(x_n)$	case and next interval	$(b - a)/2^n$
0	1.000000	2.000000	**1.500000**	–	**−0.09453**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	1.0000
1	1.500000	2.000000	**1.750000**	–	**0.30962**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.5000
2	1.500000	1.750000	**1.625000**	–	**0.11051**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.2500
3	1.500000	1.625000	**1.562500**	–	**0.00879**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.1250
4	1.500000	1.562500	**1.531250**	–	**−0.04267**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0625
5	1.531250	1.562500	**1.546875**	–	**−0.01689**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0313
6	1.546875	1.562500	**1.554688**	–	**−0.00404**	b) $f(x_n)f(b_n) < 0 \Rightarrow a_{n+1} = x_n$	0.0156
7	1.554688	1.562500	**1.558594**	–	**0.00238**	a) $f(a_n)f(x_n) < 0 \Rightarrow b_{n+1} = x_n$	0.0078

How to read a row: compute x_n, evaluate $f(x_n)$, compare its sign with $f(a_n)$; the case column gives $[a_{n+1}, b_{n+1}]$ of the next row.

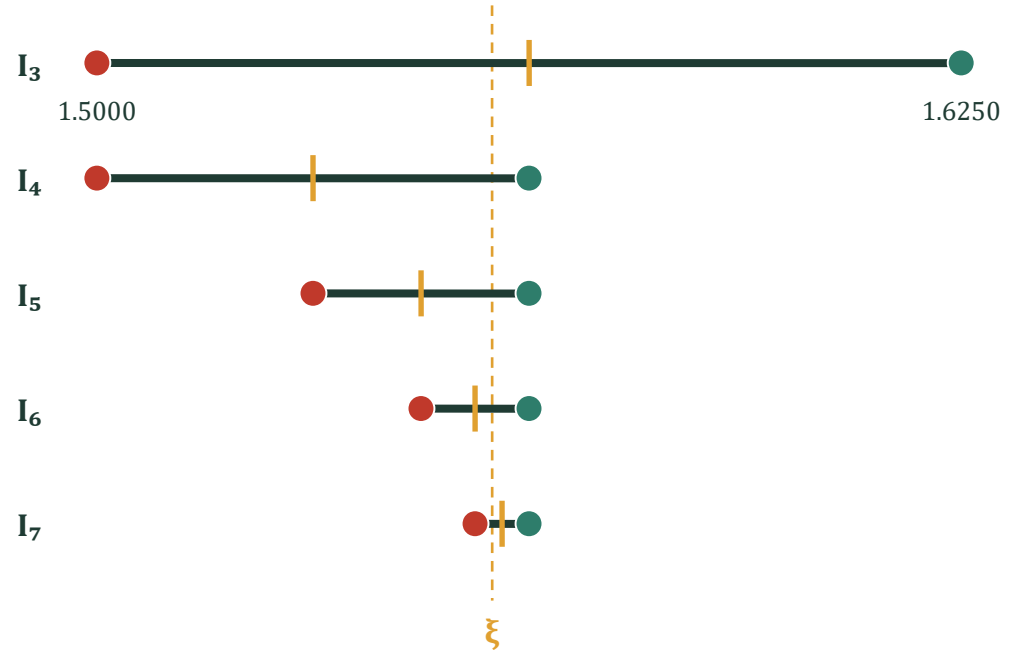
Stop at n = 7: $(b - a)/2^7 = 0.0078 \le \varepsilon = 0.01 \Rightarrow \xi \approx x_7 = 1.558594$

Example B5 · the nested intervals, drawn to scale

Full view: I_0 … I_3 on [1, 2]



Zoom ×8: I_3 … I_7 on [1.5000, 1.6250]



$\xi \approx x_7 = 1.558594$ exact $\xi = 1.55714560$

true error $|x_7 - \xi| = 1.45\times10^{-3}$ ≤ guaranteed bound 0.0078

Part II · The secant method (regula falsi)

Replace the curve by the chord through two points and take the chord's intersection with the x-axis

Hypotheses of the secant method

1 **Regularity**

$f : [a, b] \to \mathbb{R}$ twice differentiable, continuous on $[a, b]$

2 **(1) Sign change**

$f(a) \cdot f(b) < 0$

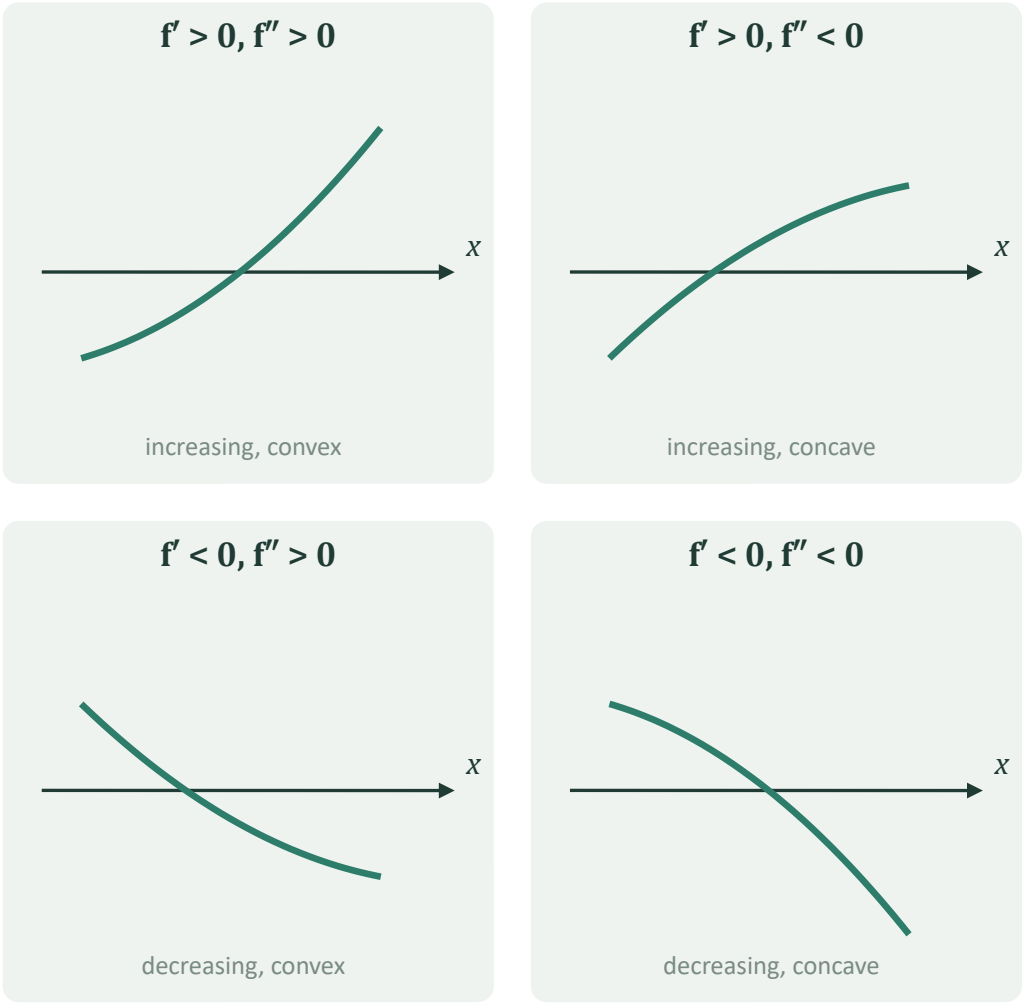
3 **(2) No turning, no bending change**

$f'(x) \cdot f''(x) \neq 0$ for every $x \in [a, b]$

Meaning of (2)

$f' \neq 0 \Rightarrow f'$ keeps one sign $\Rightarrow f$ strictly monotone $\Rightarrow$ the root ξ is unique.

$f'' \neq 0 \Rightarrow f''$ keeps one sign $\Rightarrow f$ is convex or concave on the whole interval (no inflection point).



Only two situations at the ends: the equivalences i) and ii)

i) $f(a) \cdot f''(a) < 0 \iff f(b) \cdot f''(b) > 0$

ii) $f(a) \cdot f''(a) > 0 \iff f(b) \cdot f''(b) < 0$

1 **f″ has one sign on [a, b]**

by hypothesis (2): sign $f''(a)$ = sign $f''(b)$

2 **f has opposite signs at the ends**

by hypothesis (1): sign $f(a)$ = − sign $f(b)$

3 **Multiply**

$\text{sign}(f(a)f''(a)) = -\,\text{sign}(f(b)f''(b)) \Rightarrow$ i) and ii) follow

4 **Consequence**

exactly one endpoint e satisfies $f(e) \cdot f''(e) > 0$; call it the fixed endpoint

Rule to remember

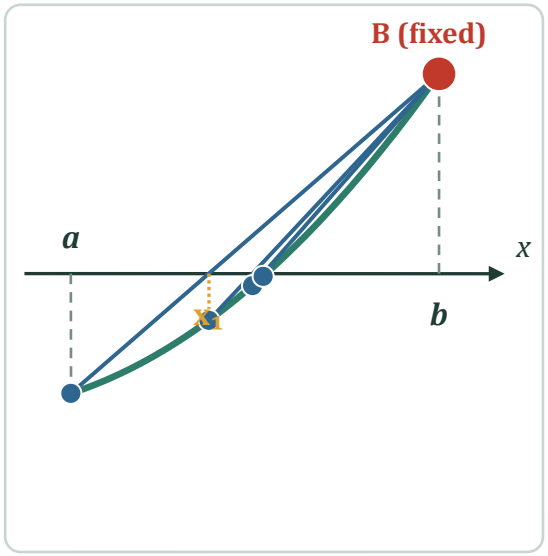
Fixed endpoint: where f and f″ have the same sign ($f \cdot f'' > 0$).

Start x_0: the other endpoint.

Of the four sign combinations at a and b, only these two remain (next slide).

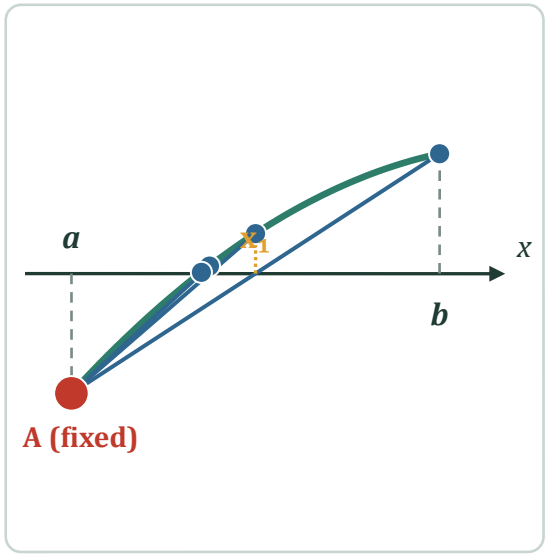
Four shapes, two rules: where the secants pivot

f′ > 0, f″ > 0



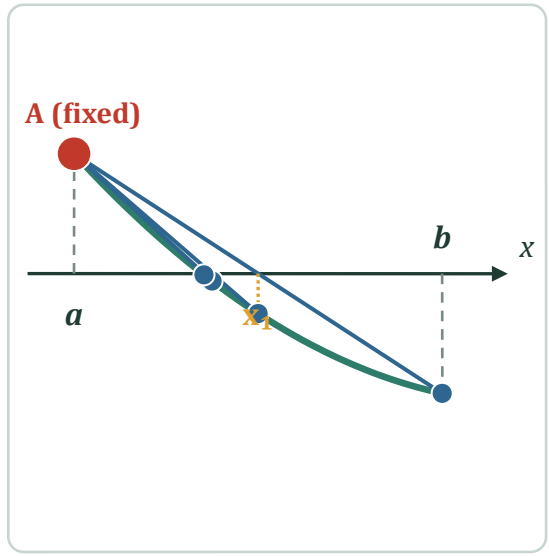
case i)
f(a)·f″(a) < 0
x_0 = a, fixed b
x_n increase → ξ

f′ > 0, f″ < 0



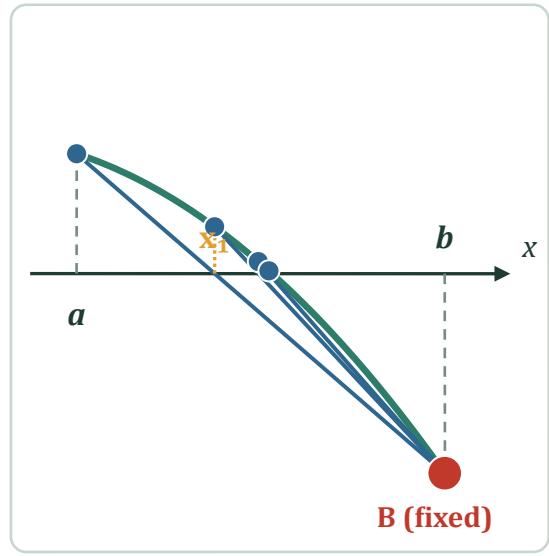
case ii)
f(b)·f″(b) < 0
x_0 = b, fixed a
x_n decrease → ξ

f′ < 0, f″ > 0



case ii)
f(b)·f″(b) < 0
x_0 = b, fixed a
x_n decrease → ξ

f′ < 0, f″ < 0



case i)
f(a)·f″(a) < 0
x_0 = a, fixed b
x_n increase → ξ

Deriving the iteration formula (case i: fixed point B)

1 **Two points**

B(b, f(b)) fixed, $A_n(x_n, f(x_n))$ current

2 **Line through them**

$(y - f(b)) / (f(x_n) - f(b)) = (x - b) / (x_n - b)$

3 **Intersect with Ox**

put y = 0: $x - b = -f(b)\cdot(x_n - b) / (f(x_n) - f(b))$

4 **Call it x_{n+1}**

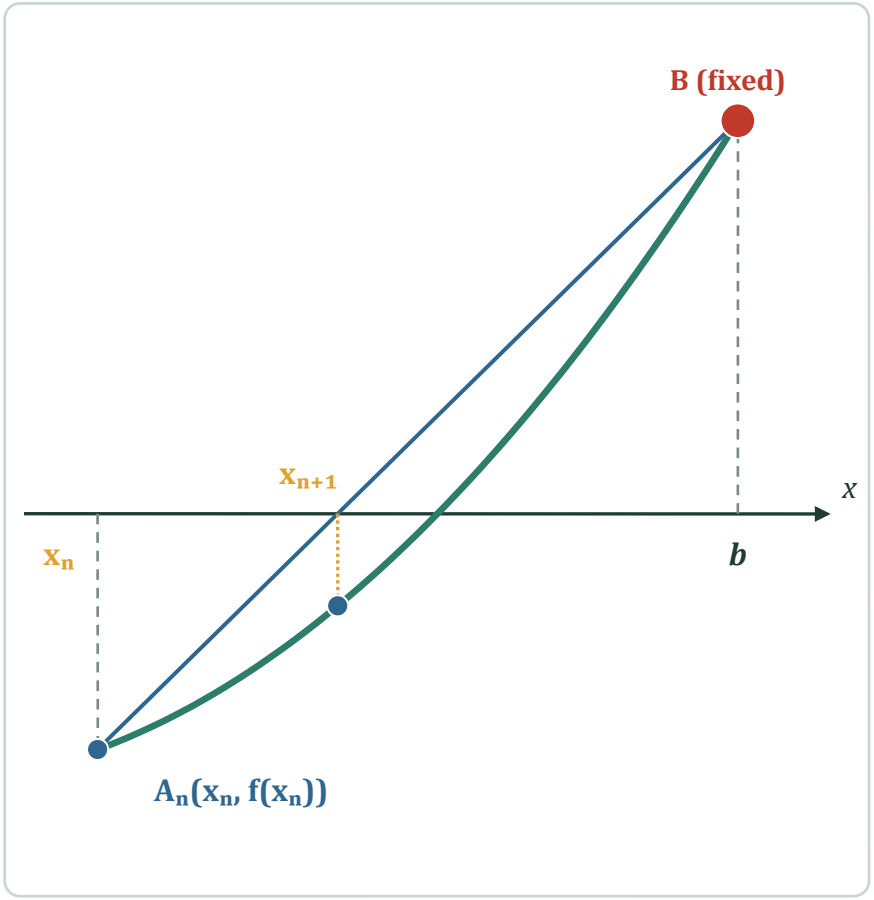
$x_{n+1} = b - f(b)(x_n - b)/(f(x_n) - f(b))$

5 **Common denominator**

$x_{n+1} = [b(f(x_n) - f(b)) - f(b)(x_n - b)] / (f(x_n) - f(b))$

$$\mathbf{x_{n+1} = (b\cdot f(x_n) - x_n\cdot f(b)) / (f(x_n) - f(b))}$$

$= x_n - f(x_n)\cdot(x_n - b)/(f(x_n) - f(b))$ (the $-b\cdot f(b)$ terms cancel)



Case ii (fixed A): replace b by a in every formula.

Theorem: the secant sequence converges monotonically to ξ

Under (1) and (2), the unique root $\xi \in (a, b)$ is the limit of a strictly monotone sequence:

i) $f(a)f''(a) < 0$: $x_0 = a$, $x_{n+1} = (b \cdot f(x_n) - x_n \cdot f(b))/(f(x_n) - f(b))$, increasing

ii) $f(b)f''(b) < 0$: $x_0 = b$, $x_{n+1} = (a \cdot f(x_n) - x_n \cdot f(a))/(f(x_n) - f(a))$, decreasing

1 Moves the right way

$f(x_n)$ and $f(b)$ have opposite signs ⇒ x_{n+1} lies strictly between x_n and b

2 Never overshoots

convexity: the chord stays on one side of the graph ⇒ x_{n+1} stays on the same side of ξ as x_n

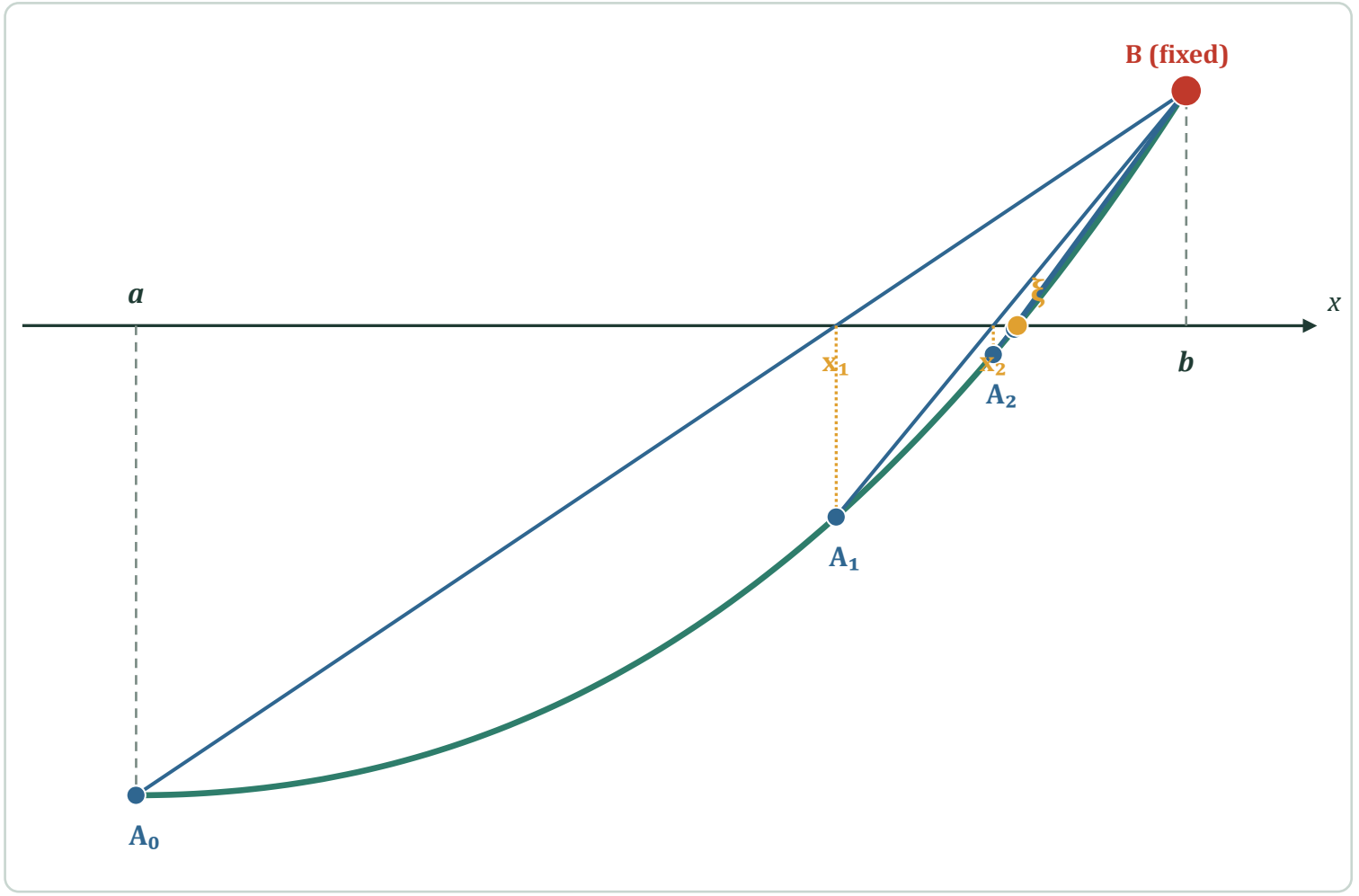
3 Monotone + bounded

case i: $x_0 < x_1 < \ldots < \xi \Rightarrow x_n \to L \leq \xi$

4 Limit is a root

pass to the limit: $L = L - f(L)(L - b)/(f(L) - f(b)) \Rightarrow f(L)(L - b) = 0 \Rightarrow f(L) = 0 \Rightarrow L = \xi$

The geometric picture: successive secants from the fixed point



Reading the picture

- Blue chords all start at B, the fixed point.
- Chord BA_n meets Ox at x_{n+1} (dotted gold line up to the curve gives A_{n+1}).
- The x_n climb towards ξ from the left and never pass it.
- Course function $x^3 - x^2 - x - 1$ on [1, 2]: case i, $f(1) < 0$, $f'' > 0$.

Observation: the same method seen as interval division

1 **Secant through the bracket ends**

$(x - a_i)/(b_i - a_i) = -f(a_i)/(f(b_i) - f(a_i))$

2 **Its zero**

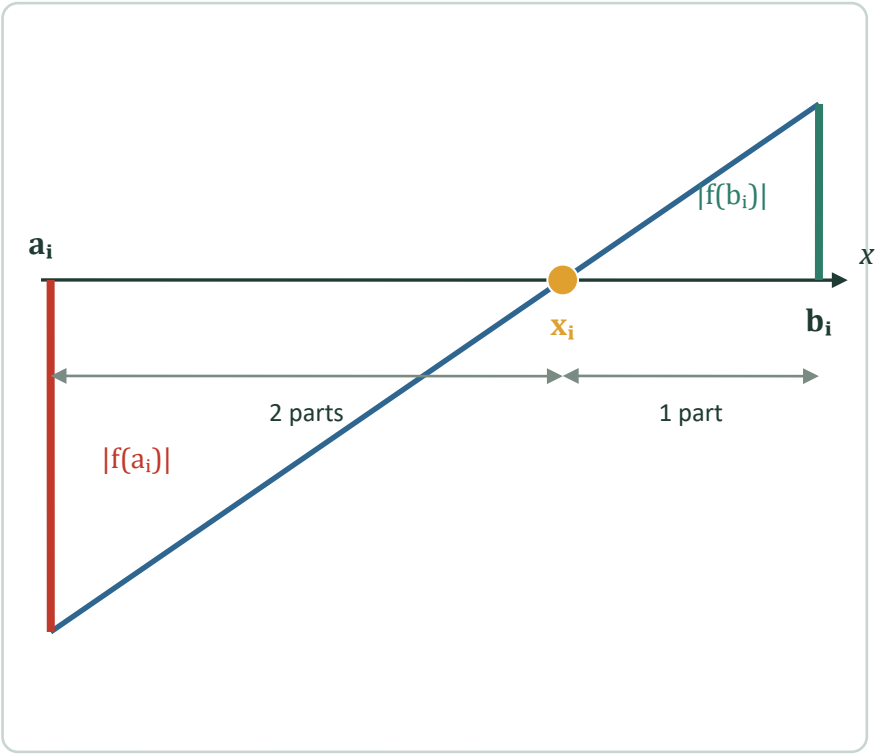
$x_i = (a_i \cdot f(b_i) - b_i \cdot f(a_i)) \,/\, (f(b_i) - f(a_i))$

3 **Division ratio**

$(x_i - a_i) : (b_i - x_i) = |f(a_i)| : |f(b_i)|$ (not halves, as in bisection)

4 **Keep the sign change**

$f(a_i)f(x_i) < 0 \Rightarrow [a_i, x_i]$; $f(a_i)f(x_i) > 0 \Rightarrow [x_i, b_i]$



Under hypothesis (2) one end never moves: it is the fixed endpoint. So this bracketing form produces exactly the same points as the formulas of the theorem; the course example is computed this way.

Similar triangles: $|f(a_i)| = 2$, $|f(b_i)| = 1 \Rightarrow x_i$ splits $[a_i, b_i]$ in ratio 2 : 1.

How accurate is x_n? Stopping criteria for the secant method

Practical criteria

$\lvert f(x_n)\rvert < \varepsilon$	used in the course example
$\lvert x_{n+1} - x_n\rvert < \varepsilon$	the iterates stop moving

A guaranteed bound (useful complement)

1 Mean value theorem

$f(x_n) - f(\xi) = f'(c)(x_n - \xi)$, c between x_n and ξ

2 Use f(ξ) = 0

$|x_n - \xi| = |f(x_n)| / |f'(c)|$

3 Bound |f′| from below

$m_1 = \min |f'(x)|$ on $[a, b] > 0$ (hypothesis 2)

$$|x_n - \xi| \leq |f(x_n)| / m_1$$

Why this matters

- A small residual $|f(x_n)|$ is an error guarantee only after dividing by m_1: if f is flat (m_1 small), a small $|f(x_n)|$ can still hide a large error.
- Course remark 2: the formula of the theorem lets us evaluate the error once x_n is computed (a-posteriori).
- The examples report all three: $|f(x_n)|$, $|x_{n+1} - x_n|$ and $|f(x_n)|/m_1$.

The secant algorithm in five steps

1	**Check**	$f(a)f(b) < 0$, and the signs of f', f'' are constant on $[a, b]$
2	**Choose**	fixed endpoint e with $f(e)\cdot f''(e) > 0$; x_0 = the other endpoint
3	**Iterate**	$x_{n+1} = (e\cdot f(x_n) - x_n\cdot f(e)) \,/\, (f(x_n) - f(e))$
4	**Test**	$\lvert f(x_{n+1})\rvert < \varepsilon$, or $\lvert x_{n+1} - x_n\rvert < \varepsilon$, or $\lvert f(x_{n+1})\rvert / m_1 < \varepsilon$
5	**Answer**	$\xi \approx x_{n+1}$ (lies on the side of ξ facing x_0)

Course observations

- The subintervals are not halved: they are split in the ratio of $|f|$ at the ends.
- Geometrically, f is replaced by the secant through $(a_i, f(a_i))$ and $(b_i, f(b_i))$.
- In general the secant method is more efficient than bisection.
- It also supplies starting values for other iterative methods (e.g. Newton).

Example S1 · secant for $x^3 - x^2 - x - 1 = 0$: hypotheses and case

$f(x) = x^3 - x^2 - x - 1$, $f'(x) = 3x^2 - 2x - 1 = (3x + 1)(x - 1)$, $f''(x) = 6x - 2$

	x = a = 1	x = b = 2
f(x)	−2.0000	1.0000
f′(x)	0	7.0000
f″(x)	4.0000	10.0000

(1) $f(a)\cdot f(b) = -2.0000 < 0$ ✓

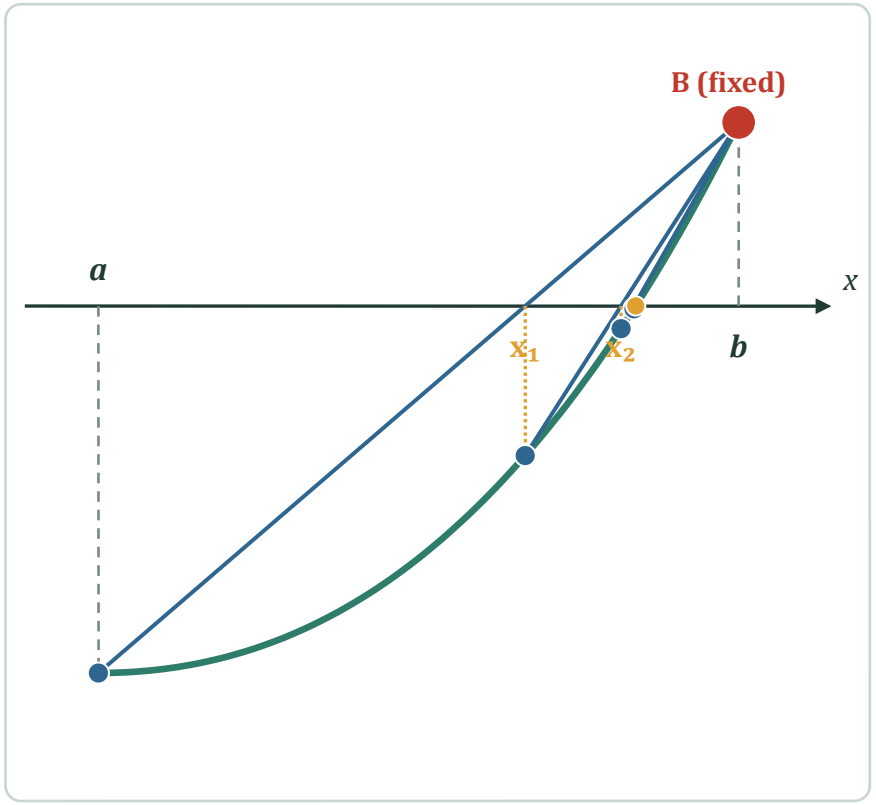
(2) $f' > 0$ on $(1, 2]$ ($f'(1) = 0$ only at the end); $f'' \geq 4 > 0$ on $[1, 2]$ ✓

case $f(a)\cdot f''(a) = -8.0000 < 0 \Rightarrow$ case i)

setup $x_0 = a = 1$, fixed endpoint $b = 2$, $f(b) = 1.0000$

$$x_{n+1} = (2\cdot f(x_n) - x_n\cdot(1.0000)) \,/\, (f(x_n) - (1.0000))$$

Pivot B = (b, f(b)); x_n increase towards ξ



f′(1) = 0 at the left end only: (2) holds on (1, 2]; the method still works, and m_1 is taken on [5/3, 2].

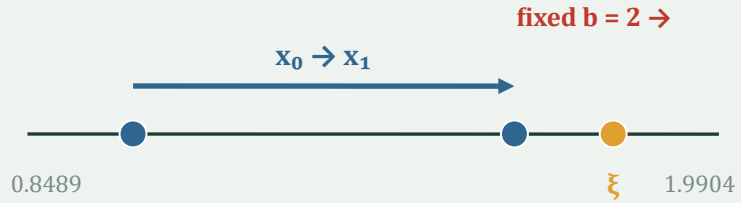
Example S1 · the first iterations, step by step

1

$x_0 = 1$, $f(x_0) = 1 - 1 - 1 - 1 = -2$

$x_1 = (2\cdot(-2) - 1\cdot 1) / (-2 - 1) = -5 / -3 = 5/3 \approx 1.666667$ (course: "x_0")

$f(x_1) = 125/27 - 25/9 - 5/3 - 1 = -22/27 \approx -0.8148$ ($|f| \geq 0.2$: continue)

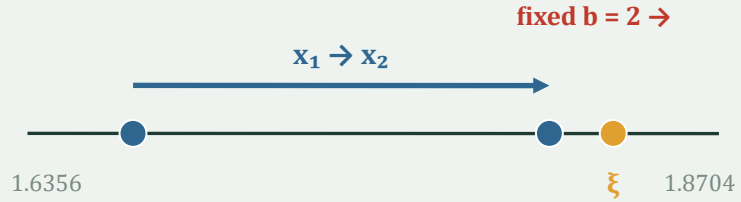


2

$x_1 = 5/3$, $f(x_1) = -22/27$

$x_2 = (2\cdot(-22/27) - (5/3)\cdot 1) / (-22/27 - 1) = (-89/27)/(-49/27) = 89/49 \approx 1.816327$ (course: "x_1")

$f(x_2) = -14498/117649 \approx -0.1232$ ⇒ $|f(x_2)| < 0.2$: course stops here

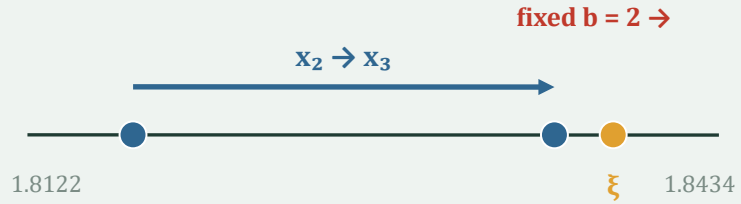


3

$x_2 = 1.816327$, $f(x_2) = -0.123231$

$x_3 = (2\cdot(-0.123231) - 1.816327\cdot(1.000000)) / ((-0.123231) - (1.000000)) = 1.836478$

$f(x_3) = -0.015332$, $|x_3 - x_2| = 0.020151$

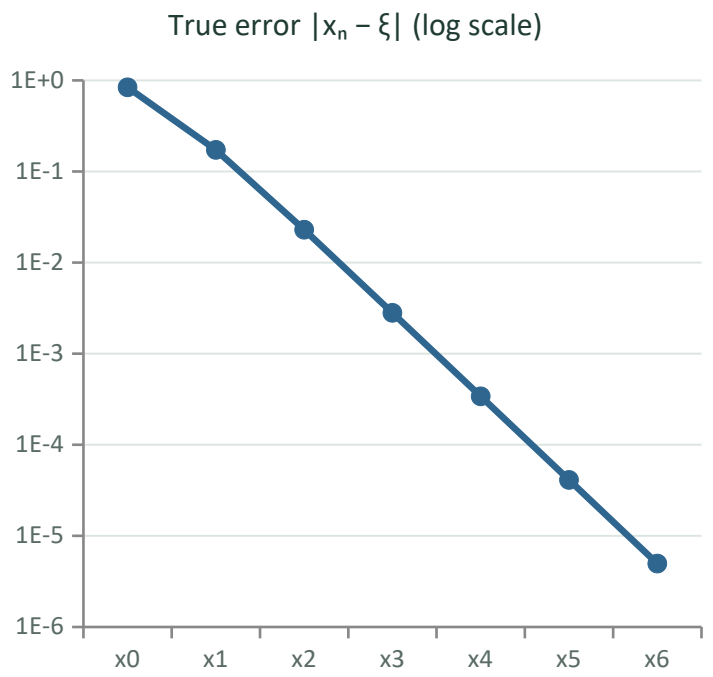


Right: each step on a number line zoomed around x_n, x_{n+1} and ξ (gold); the fixed endpoint b lies outside, in the direction shown.

Example S1 · all iterations, error control and result

n	x_n	$f(x_n)$	$\lvert x_{n+1} - x_n \rvert$	$\lvert f(x_n) \rvert / m_1$	true $\lvert x_n - \xi \rvert$
0	1.00000000	-2.00×10^{0}	6.67×10^{-1}	—	8.39×10^{-1}
1	1.66666667	-8.15×10^{-1}	1.50×10^{-1}	2.04×10^{-1}	1.73×10^{-1}
2	1.81632653	-1.23×10^{-1}	2.02×10^{-2}	3.08×10^{-2}	2.30×10^{-2}
3	1.83647756	-1.53×10^{-2}	2.47×10^{-3}	3.83×10^{-3}	2.81×10^{-3}
4	1.83894677	-1.86×10^{-3}	2.99×10^{-4}	4.65×10^{-4}	3.40×10^{-4}
5	1.83924566	-2.25×10^{-4}	3.61×10^{-5}	5.62×10^{-5}	4.11×10^{-5}
6	1.83928179	-2.72×10^{-5}	4.37×10^{-6}	6.79×10^{-6}	4.97×10^{-6}

$m_1 = \min |f'|$ on $[1.6667, 2.0000] = 4.0000$



$\xi \approx x_5 = 1.839246$ ($|f(x_5)|/m_1 = 5.62\times10^{-5} < 10^{-4}$) exact $\xi = 1.83928676$

For the same guarantee 10^{-4}, bisection needs 14 steps; the secant method needs 5.

Example S2 · secant for $x^3 + x - 1 = 0$: hypotheses and case

$f(x) = x^3 + x - 1$, $f'(x) = 3x^2 + 1$, $f''(x) = 6x$

	x = a = 0.5	x = b = 1
f(x)	−0.3750	1.0000
f′(x)	1.7500	4.0000
f″(x)	3.0000	6.0000

(1) $f(a)\cdot f(b) = -0.3750 < 0$ ✓

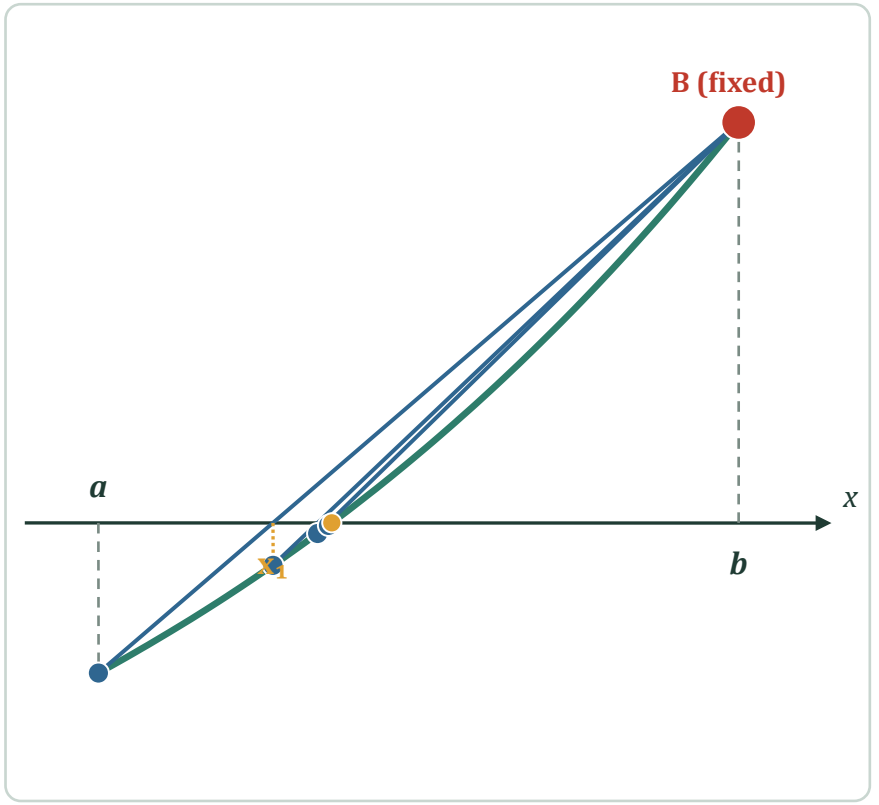
(2) $f' \geq 1.75 > 0$ on $[0.5, 1]$; $f'' \geq 3 > 0$ on $[0.5, 1]$ ✓

case $f(a)\cdot f''(a) = -1.1250 < 0 \Rightarrow$ case i)

setup $x_0 = a = 0.5$, fixed endpoint $b = 1$, $f(b) = 1.0000$

$$x_{n+1} = (1\cdot f(x_n) - x_n\cdot(1.0000)) \,/\, (f(x_n) - (1.0000))$$

Pivot B = (b, f(b)); x_n increase towards ξ



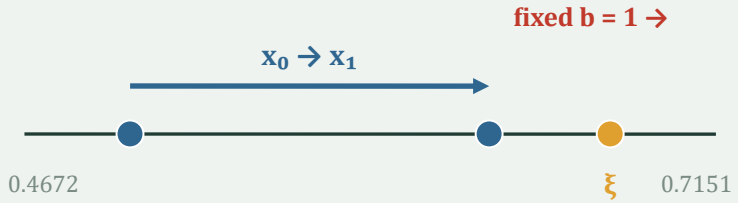
Example S2 · the first iterations, step by step

1

$x_0 = 0.500000$, $f(x_0) = -0.375000$

$$\mathbf{x_1 = (1\cdot(-0.375000) - 0.500000\cdot(1.000000)) / ((-0.375000) - (1.000000)) = 0.636364}$$

$f(x_1) = -0.105935$, $|x_1 - x_0| = 0.136364$

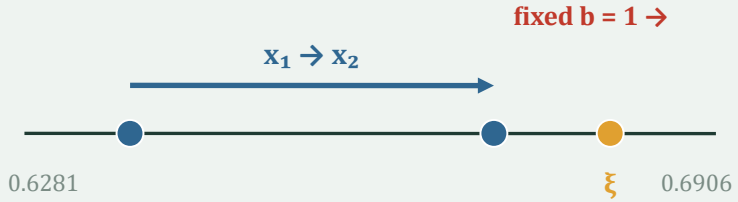


2

$x_1 = 0.636364$, $f(x_1) = -0.105935$

$$\mathbf{x_2 = (1\cdot(-0.105935) - 0.636364\cdot(1.000000)) / ((-0.105935) - (1.000000)) = 0.671196}$$

$f(x_2) = -0.026428$, $|x_2 - x_1| = 0.034832$

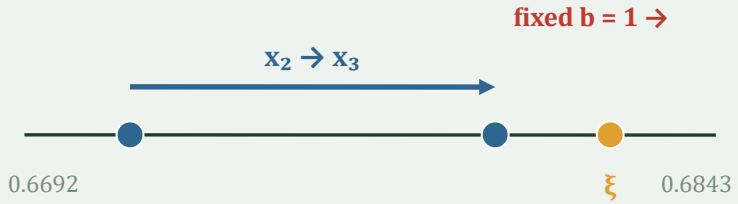


3

$x_2 = 0.671196$, $f(x_2) = -0.026428$

$$\mathbf{x_3 = (1\cdot(-0.026428) - 0.671196\cdot(1.000000)) / ((-0.026428) - (1.000000)) = 0.679662}$$

$f(x_3) = -0.006375$, $|x_3 - x_2| = 0.008466$

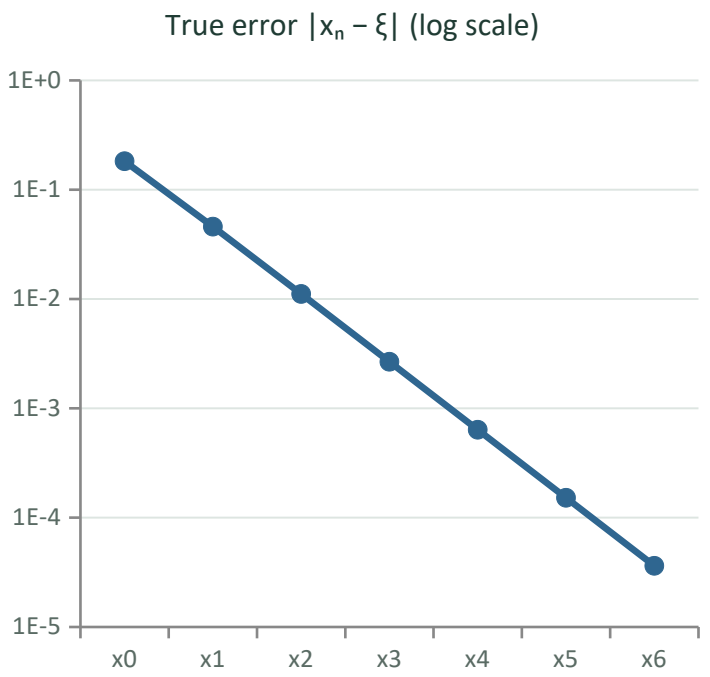


Right: each step on a number line zoomed around x_n, x_{n+1} and ξ (gold); the fixed endpoint b lies outside, in the direction shown.

Example S2 · all iterations, error control and result

n	x_n	$f(x_n)$	$\lvert x_{n+1} - x_n \rvert$	$\lvert f(x_n) \rvert / m_1$	true $\lvert x_n - \xi \rvert$
0	0.50000000	-3.75×10^{-1}	1.36×10^{-1}	2.14×10^{-1}	1.82×10^{-1}
1	0.63636364	-1.06×10^{-1}	3.48×10^{-2}	6.05×10^{-2}	4.60×10^{-2}
2	0.67119565	-2.64×10^{-2}	8.47×10^{-3}	1.51×10^{-2}	1.11×10^{-2}
3	0.67966165	-6.38×10^{-3}	2.03×10^{-3}	3.64×10^{-3}	2.67×10^{-3}
4	0.68169102	-1.53×10^{-3}	4.85×10^{-4}	8.72×10^{-4}	6.37×10^{-4}
5	0.68217582	-3.64×10^{-4}	1.16×10^{-4}	2.08×10^{-4}	1.52×10^{-4}
6	0.68229153	-8.69×10^{-5}	2.76×10^{-5}	4.97×10^{-5}	3.63×10^{-5}

$m_1 = \min |f'|$ on $[0.5000, 1.0000] = 1.7500$



$\xi \approx x_6 = 0.682292$ $(|f(x_6)|/m_1 = 4.97\times10^{-5} < 10^{-4})$ exact $\xi = 0.68232780$

For the same guarantee 10^{-4}, bisection needs 13 steps; the secant method needs 6.

Example S3 · secant for cos x − x = 0: hypotheses and case

$f(x) = \cos x - x$, $f'(x) = -\sin x - 1$, $f''(x) = -\cos x$

	x = a = 0	x = b = 1
f(x)	1.0000	−0.4597
f′(x)	−1.0000	−1.8415
f″(x)	−1.0000	−0.5403

(1) $f(a)\cdot f(b) = -0.4597 < 0$ ✓

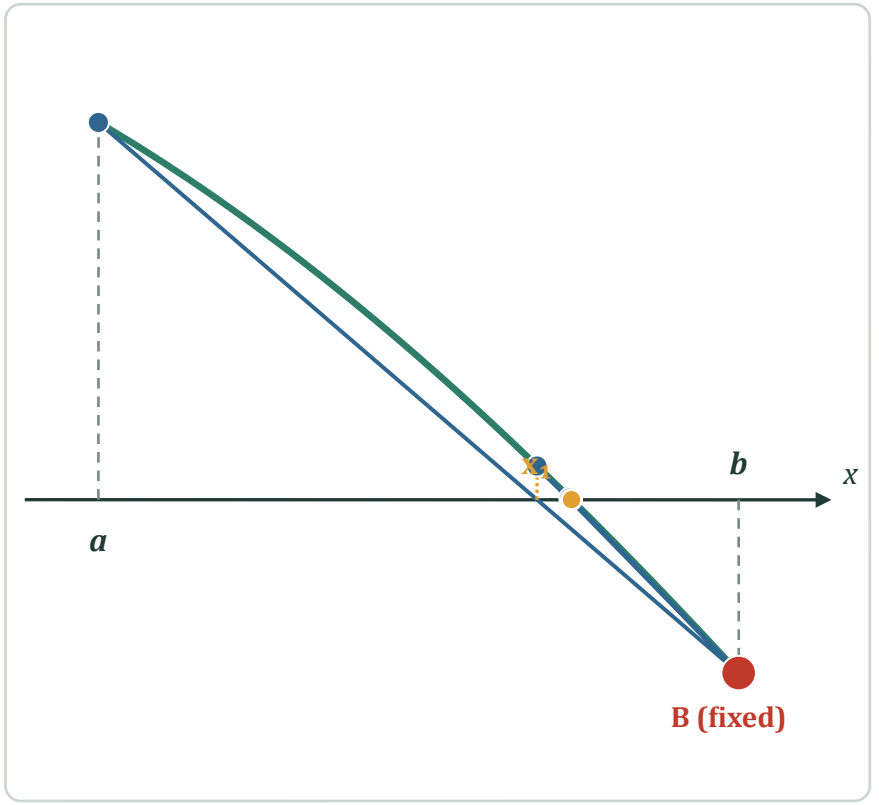
(2) $f' \le -1 < 0$ on $[0, 1]$; $f'' \le -\cos 1 < 0$ on $[0, 1]$ ✓

case $f(a)\cdot f''(a) = -1.0000 < 0 \Rightarrow$ case i)

setup $x_0 = a = 0$, fixed endpoint $b = 1$, $f(b) = -0.4597$

$$x_{n+1} = (1\cdot f(x_n) - x_n\cdot(-0.4597)) / (f(x_n) - (-0.4597))$$

Pivot B = (b, f(b)); x_n increase towards ξ



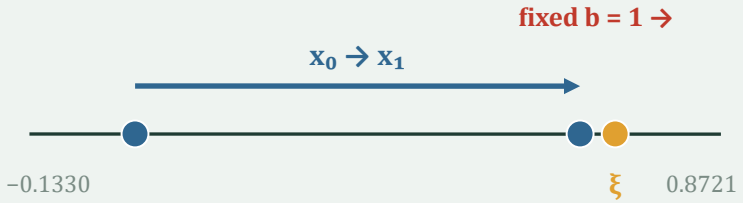
Example S3 · the first iterations, step by step

1

$x_0 = 0.000000$, $f(x_0) = 1.000000$

$$\mathbf{x_1 = (1\cdot(1.000000) - 0.000000\cdot(-0.459698)) / ((1.000000) - (-0.459698)) = 0.685073}$$

$f(x_1) = 0.089299$, $|x_1 - x_0| = 0.685073$

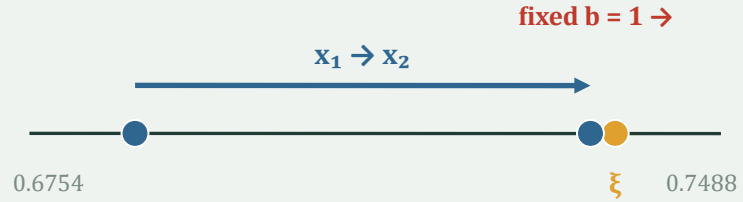


2

$x_1 = 0.685073$, $f(x_1) = 0.089299$

$$\mathbf{x_2 = (1\cdot(0.089299) - 0.685073\cdot(-0.459698)) / ((0.089299) - (-0.459698)) = 0.736299}$$

$f(x_2) = 0.004660$, $|x_2 - x_1| = 0.051226$

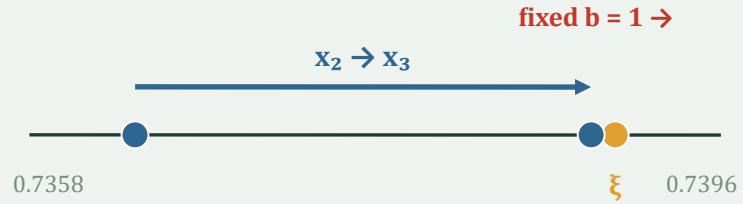


3

$x_2 = 0.736299$, $f(x_2) = 0.004660$

$$\mathbf{x_3 = (1\cdot(0.004660) - 0.736299\cdot(-0.459698)) / ((0.004660) - (-0.459698)) = 0.738945}$$

$f(x_3) = 0.000234$, $|x_3 - x_2| = 0.002646$

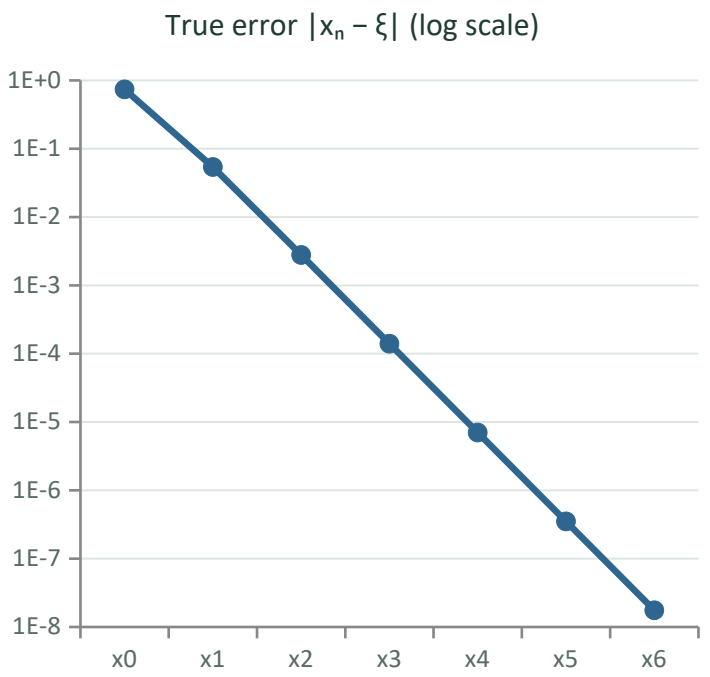


Right: each step on a number line zoomed around x_n, x_{n+1} and ξ (gold); the fixed endpoint b lies outside, in the direction shown.

Example S3 · all iterations, error control and result

n	x_n	$f(x_n)$	$\lvert x_{n+1} - x_n \rvert$	$\lvert f(x_n) \rvert / m_1$	true $\lvert x_n - \xi \rvert$
0	0.00000000	1.00×10^{0}	6.85×10^{-1}	1.00×10^{0}	7.39×10^{-1}
1	0.68507336	8.93×10^{-2}	5.12×10^{-2}	8.93×10^{-2}	5.40×10^{-2}
2	0.73629900	4.66×10^{-3}	2.65×10^{-3}	4.66×10^{-3}	2.79×10^{-3}
3	0.73894536	2.34×10^{-4}	1.33×10^{-4}	2.34×10^{-4}	1.40×10^{-4}
4	0.73907813	1.17×10^{-5}	6.65×10^{-6}	1.17×10^{-5}	7.00×10^{-6}
5	0.73908478	5.87×10^{-7}	3.33×10^{-7}	5.87×10^{-7}	3.51×10^{-7}
6	0.73908512	2.94×10^{-8}	1.67×10^{-8}	2.94×10^{-8}	1.76×10^{-8}

$m_1 = \min |f'|$ on $[0.0000, 1.0000] = 1.0000$



$\xi \approx x_4 = 0.739078$ $(|f(x_4)|/m_1 = 1.17\times10^{-5} < 10^{-4})$ exact $\xi = 0.73908513$

For the same guarantee 10^{-4}, bisection needs 14 steps; the secant method needs 4.

Example S4 · secant for $e^{-x} - x = 0$: hypotheses and case

$f(x) = e^{-x} - x$, $f'(x) = -e^{-x} - 1$, $f''(x) = e^{-x}$

	x = a = 0	x = b = 1
f(x)	1.0000	−0.6321
f′(x)	−2.0000	−1.3679
f″(x)	1.0000	0.3679

(1) $f(a)\cdot f(b) = -0.6321 < 0$ ✓

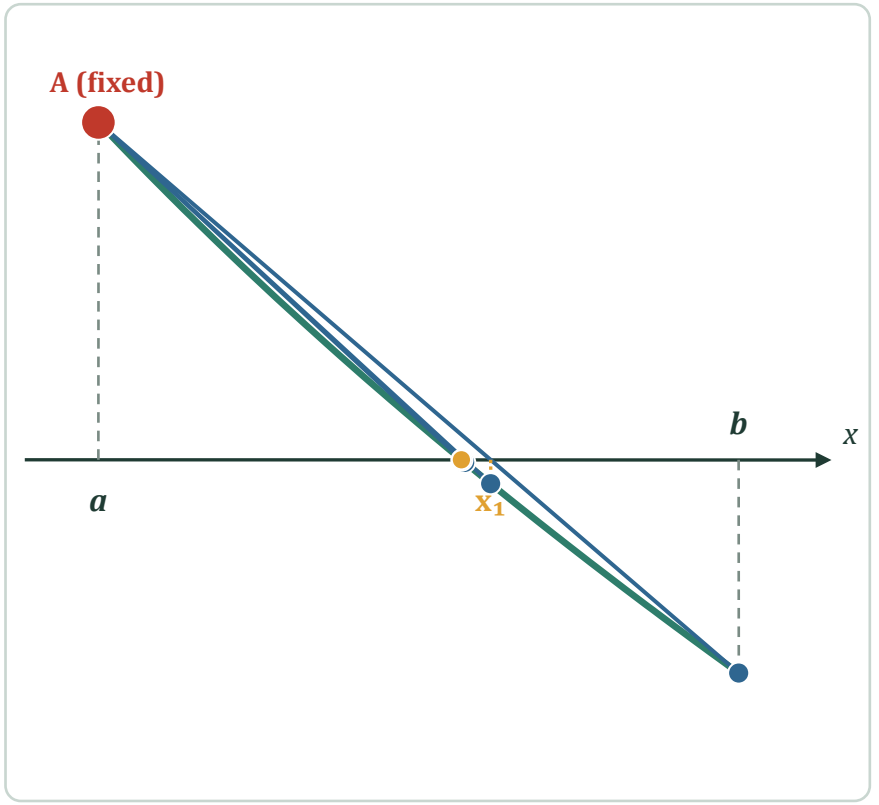
(2) $f' < -1 < 0$ on $[0, 1]$; $f'' > 0$ on $[0, 1]$ ✓

case $f(a)\cdot f''(a) = 1.0000 > 0 \Rightarrow$ case ii)

setup $x_0 = b = 1$, fixed endpoint $a = 0$, $f(a) = 1.0000$

$$x_{n+1} = (0\cdot f(x_n) - x_n\cdot(1.0000)) / (f(x_n) - (1.0000))$$

Pivot A = (a, f(a)); x_n decrease towards ξ



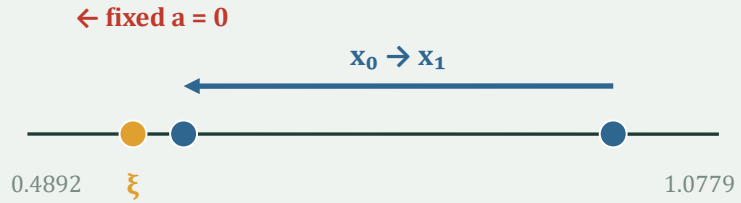
Example S4 · the first iterations, step by step

1

$x_0 = 1.000000$, $f(x_0) = -0.632121$

$$\mathbf{x_1 = (0\cdot(-0.632121) - 1.000000\cdot(1.000000)) / ((-0.632121) - (1.000000)) = 0.612700}$$

$f(x_1) = -0.070814$, $|x_1 - x_0| = 0.387300$

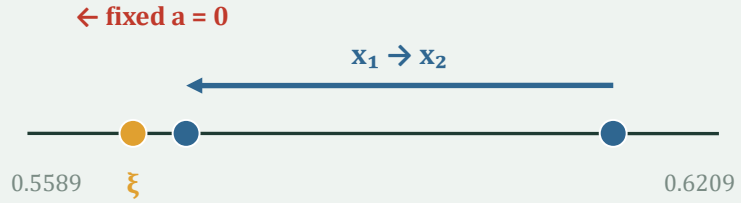


2

$x_1 = 0.612700$, $f(x_1) = -0.070814$

$$\mathbf{x_2 = (0\cdot(-0.070814) - 0.612700\cdot(1.000000)) / ((-0.070814) - (1.000000)) = 0.572181}$$

$f(x_2) = -0.007888$, $|x_2 - x_1| = 0.040518$

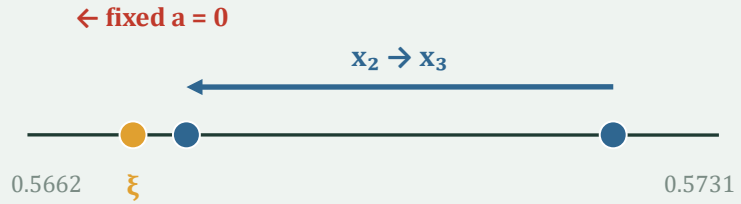


3

$x_2 = 0.572181$, $f(x_2) = -0.007888$

$$\mathbf{x_3 = (0\cdot(-0.007888) - 0.572181\cdot(1.000000)) / ((-0.007888) - (1.000000)) = 0.567703}$$

$f(x_3) = -0.000877$, $|x_3 - x_2| = 0.004478$

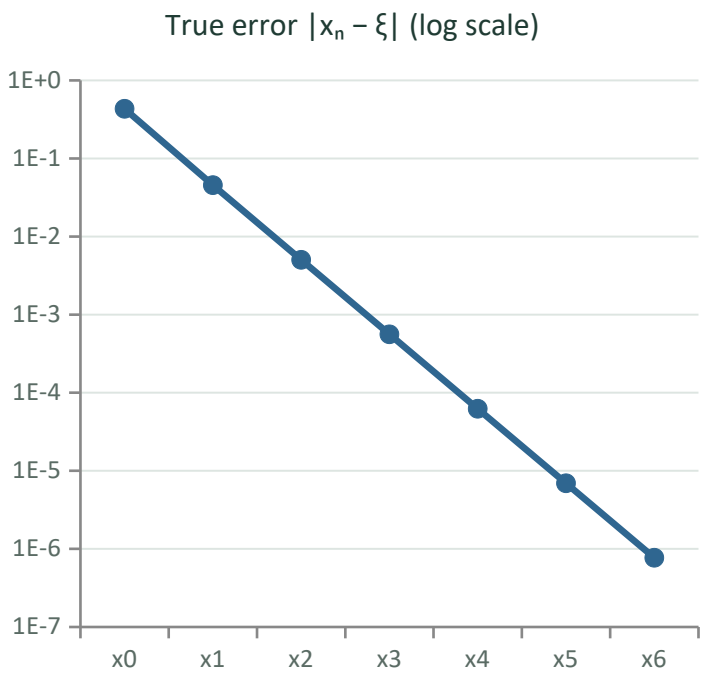


Right: each step on a number line zoomed around x_n, x_{n+1} and ξ (gold); the fixed endpoint a lies outside, in the direction shown.

Example S4 · all iterations, error control and result

n	x_n	$f(x_n)$	$\lvert x_{n+1} - x_n \rvert$	$\lvert f(x_n) \rvert / m_1$	true $\lvert x_n - \xi \rvert$
0	1.00000000	-6.32×10^{-1}	3.87×10^{-1}	4.62×10^{-1}	4.33×10^{-1}
1	0.61269984	-7.08×10^{-2}	4.05×10^{-2}	5.18×10^{-2}	4.56×10^{-2}
2	0.57218141	-7.89×10^{-3}	4.48×10^{-3}	5.77×10^{-3}	5.04×10^{-3}
3	0.56770321	-8.77×10^{-4}	4.98×10^{-4}	6.41×10^{-4}	5.60×10^{-4}
4	0.56720555	-9.76×10^{-5}	5.53×10^{-5}	7.13×10^{-5}	6.23×10^{-5}
5	0.56715021	-1.09×10^{-5}	6.15×10^{-6}	7.93×10^{-6}	6.92×10^{-6}
6	0.56714406	-1.21×10^{-6}	6.84×10^{-7}	8.82×10^{-7}	7.70×10^{-7}

$m_1 = \min |f'|$ on $[0.0000, 1.0000] = 1.3679$



$\xi \approx x_4 = 0.567206$ $(|f(x_4)|/m_1 = 7.13\times10^{-5} < 10^{-4})$ exact $\xi = 0.56714329$

For the same guarantee 10^{-4}, bisection needs 14 steps; the secant method needs 4.

Example S5 · secant for ln x + x − 2 = 0: hypotheses and case

$f(x) = \ln x + x - 2$, $f'(x) = 1/x + 1$, $f''(x) = -1/x^2$

	x = a = 1	x = b = 2
f(x)	−1.0000	0.6931
f′(x)	2.0000	1.5000
f″(x)	−1.0000	−0.2500

(1) $f(a)\cdot f(b) = -0.6931 < 0$ ✓

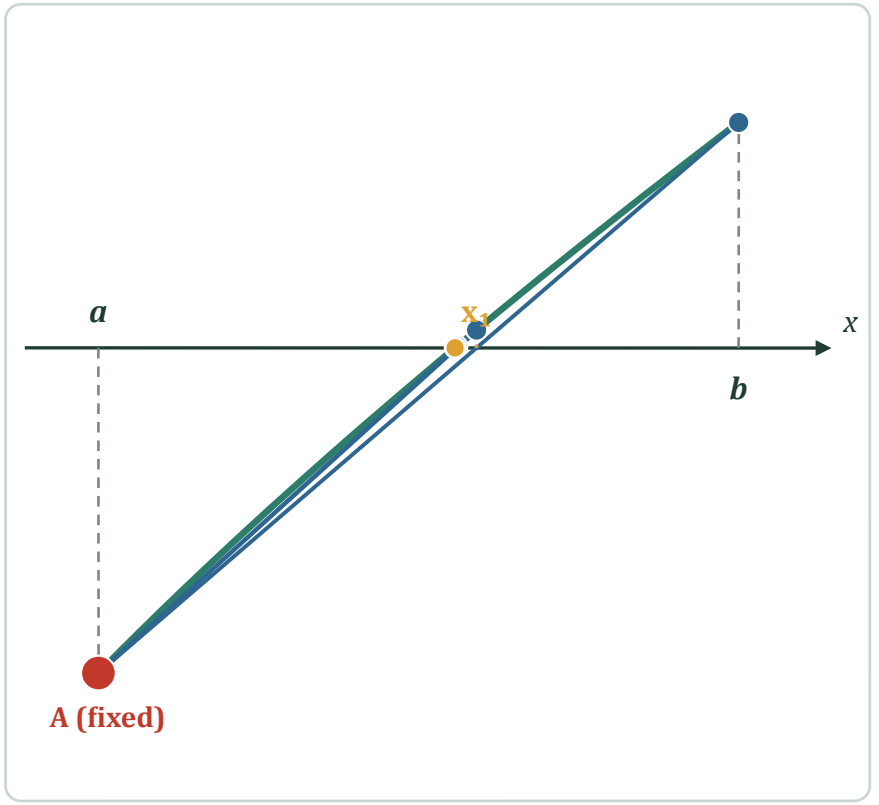
(2) $f' > 1 > 0$ on $[1, 2]$; $f'' < 0$ on $[1, 2]$ ✓

case $f(a)\cdot f''(a) = 1.0000 > 0 \Rightarrow$ case ii)

setup $x_0 = b = 2$, fixed endpoint $a = 1$, $f(a) = -1.0000$

$$x_{n+1} = (1\cdot f(x_n) - x_n\cdot(-1.0000)) / (f(x_n) - (-1.0000))$$

Pivot A = (a, f(a)); x_n decrease towards ξ



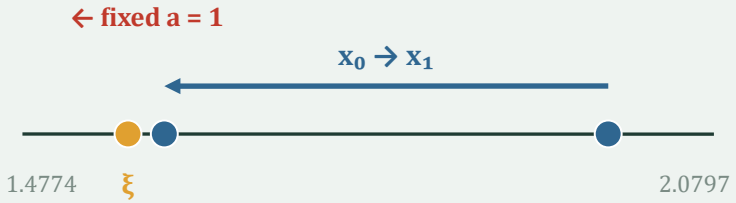
Example S5 · the first iterations, step by step

1

$x_0 = 2.000000$, $f(x_0) = 0.693147$

$$\mathbf{x_1 = (1\cdot(0.693147) - 2.000000\cdot(-1.000000)) / ((0.693147) - (-1.000000)) = 1.590616}$$

$f(x_1) = 0.054738$, $|x_1 - x_0| = 0.409384$

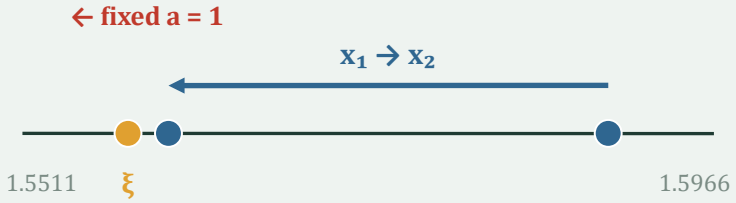


2

$x_1 = 1.590616$, $f(x_1) = 0.054738$

$$\mathbf{x_2 = (1\cdot(0.054738) - 1.590616\cdot(-1.000000)) / ((0.054738) - (-1.000000)) = 1.559965}$$

$f(x_2) = 0.004628$, $|x_2 - x_1| = 0.030651$

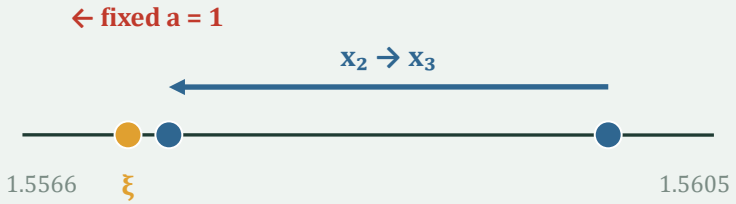


3

$x_2 = 1.559965$, $f(x_2) = 0.004628$

$$\mathbf{x_3 = (1\cdot(0.004628) - 1.559965\cdot(-1.000000)) / ((0.004628) - (-1.000000)) = 1.557385}$$

$f(x_3) = 0.000393$, $|x_3 - x_2| = 0.002580$

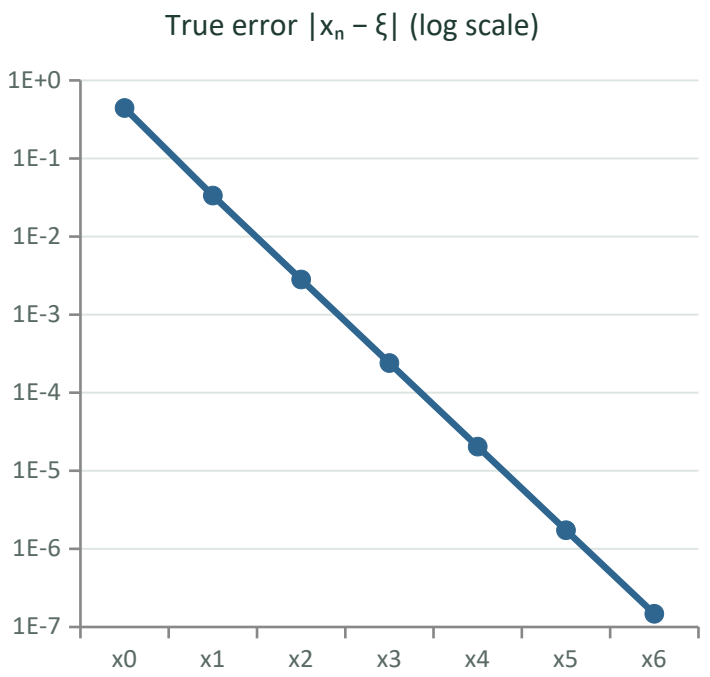


Right: each step on a number line zoomed around x_n, x_{n+1} and ξ (gold); the fixed endpoint a lies outside, in the direction shown.

Example S5 · all iterations, error control and result

n	x_n	$f(x_n)$	$\lvert x_{n+1} - x_n \rvert$	$\lvert f(x_n) \rvert / m_1$	true $\lvert x_n - \xi \rvert$
0	2.00000000	6.93×10^{-1}	4.09×10^{-1}	4.62×10^{-1}	4.43×10^{-1}
1	1.59061611	5.47×10^{-2}	3.07×10^{-2}	3.65×10^{-2}	3.35×10^{-2}
2	1.55996500	4.63×10^{-3}	2.58×10^{-3}	3.09×10^{-3}	2.82×10^{-3}
3	1.55738521	3.93×10^{-4}	2.19×10^{-4}	2.62×10^{-4}	2.40×10^{-4}
4	1.55716598	3.35×10^{-5}	1.86×10^{-5}	2.23×10^{-5}	2.04×10^{-5}
5	1.55714733	2.85×10^{-6}	1.59×10^{-6}	1.90×10^{-6}	1.73×10^{-6}
6	1.55714575	2.42×10^{-7}	1.35×10^{-7}	1.61×10^{-7}	1.47×10^{-7}

$m_1 = \min |f'|$ on $[1.0000, 2.0000] = 1.5000$



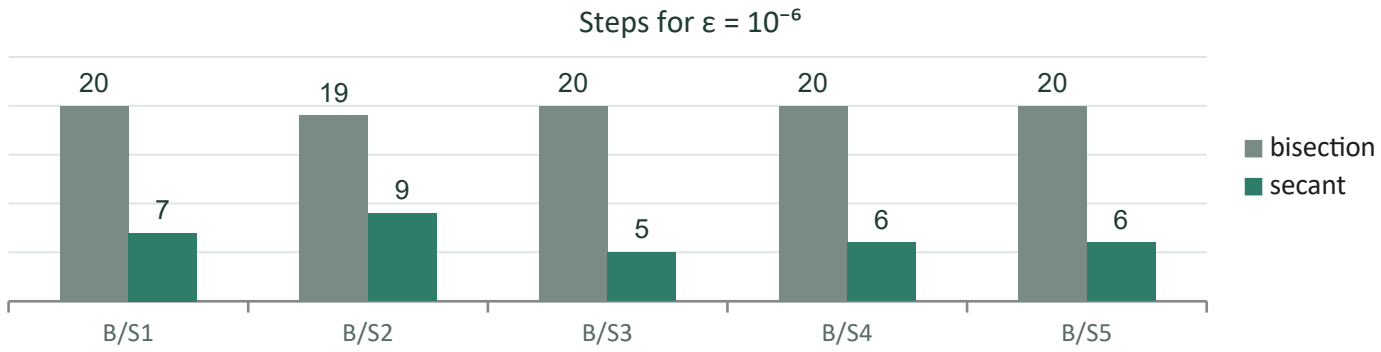
$\xi \approx x_4 = 1.557166$ $(|f(x_4)|/m_1 = 2.23\times10^{-5} < 10^{-4})$ exact $\xi = 1.55714560$

For the same guarantee 10^{-4}, bisection needs 14 steps; the secant method needs 4.

Bisection vs secant on the same five equations

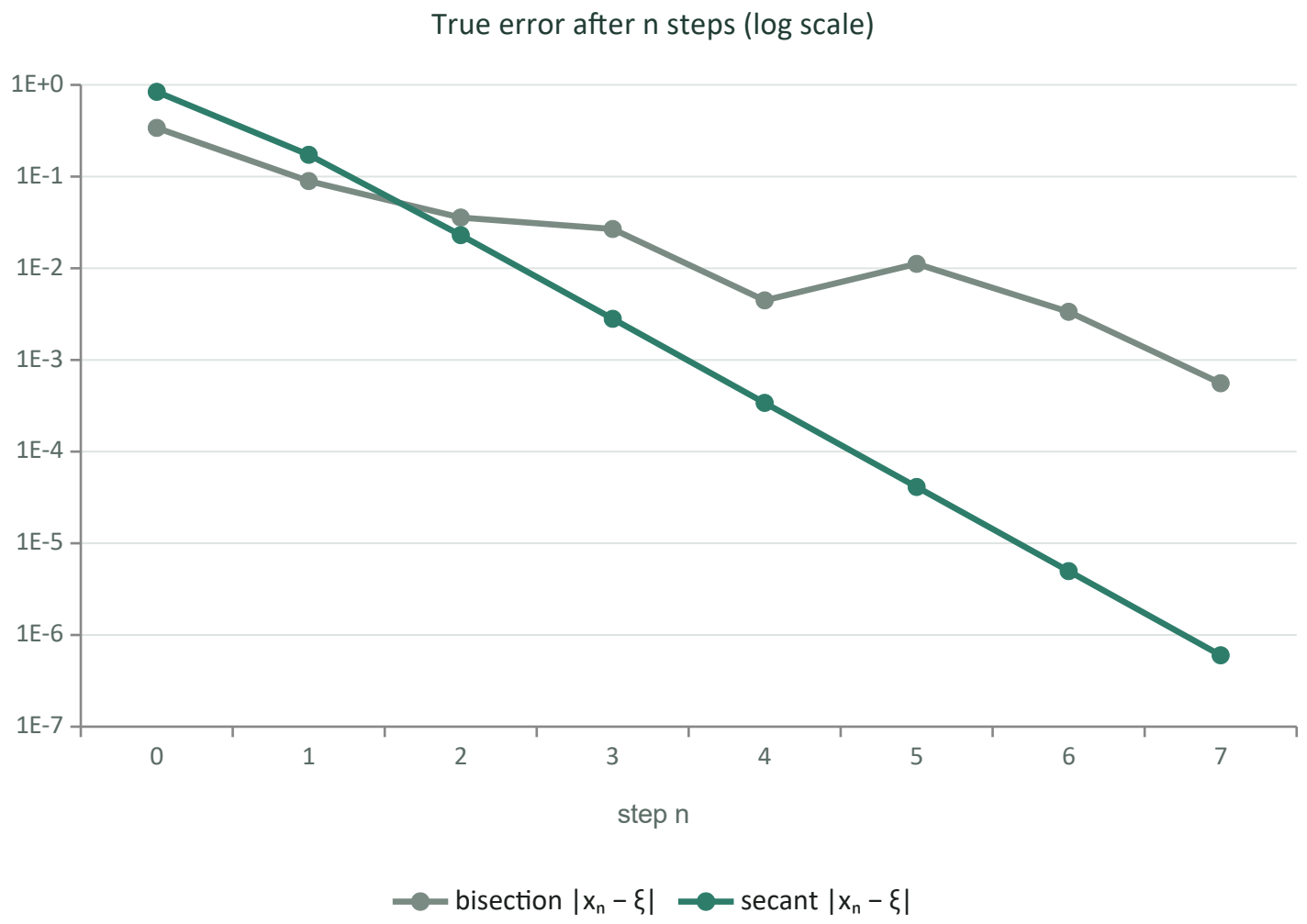
equation	[a, b]	root ξ	bis 10^{-2}	sec 10^{-2}	bis 10^{-4}	sec 10^{-4}	bis 10^{-6}	sec 10^{-6}
$x^3 - x^2 - x - 1$	[1, 2]	1.839287	7	**3**	14	**5**	20	**7**
$x^3 + x - 1$	[0.5, 1]	0.682328	6	**3**	13	**6**	19	**9**
$\cos x - x$	[0, 1]	0.739085	7	**2**	14	**4**	20	**5**
$e^{-x} - x$	[0, 1]	0.567143	7	**2**	14	**4**	20	**6**
$\ln x + x - 2$	[1, 2]	1.557146	7	**2**	14	**4**	20	**6**

Steps until the guaranteed error is below ε: bisection $(b - a)/2^n \le \varepsilon$; secant $|f(x_n)|/m_1 < \varepsilon$.



Bisection: about 3.3 steps per decimal, whatever f is. Secant: uses the values of f, so each step gains roughly one decimal or more on these examples.

Error history on the course equation $x^3 - x^2 - x - 1 = 0$



What the chart shows

- Bisection: the error drops irregularly (the midpoint may land close to ξ by chance), but its bound halves each step.
- Secant: the error shrinks by a nearly constant factor (≈ 0.12 here) every step, a straight line on the log scale.
- After 7 steps: bisection 5.57×10^{-4}, secant 6.00×10^{-7}.

Key takeaways

1 Start

Both methods need a separated root: [a, b] with $f(a)\cdot f(b) < 0$ and exactly one root inside (Rolle's sequence).

2 Bisection

Halve the bracket, keep the half with the sign change. Always converges; error $\le (b - a)/2^n$ known in advance; about 3.3 steps per decimal.

3 Secant

Pivot chords on the endpoint where $f\cdot f'' > 0$; start from the other end. Monotone convergence, far fewer steps; error $\le |f(x_n)|/m_1$.

In practice: a few bisection steps for a safe bracket, then a faster method (secant, Newton) to finish.